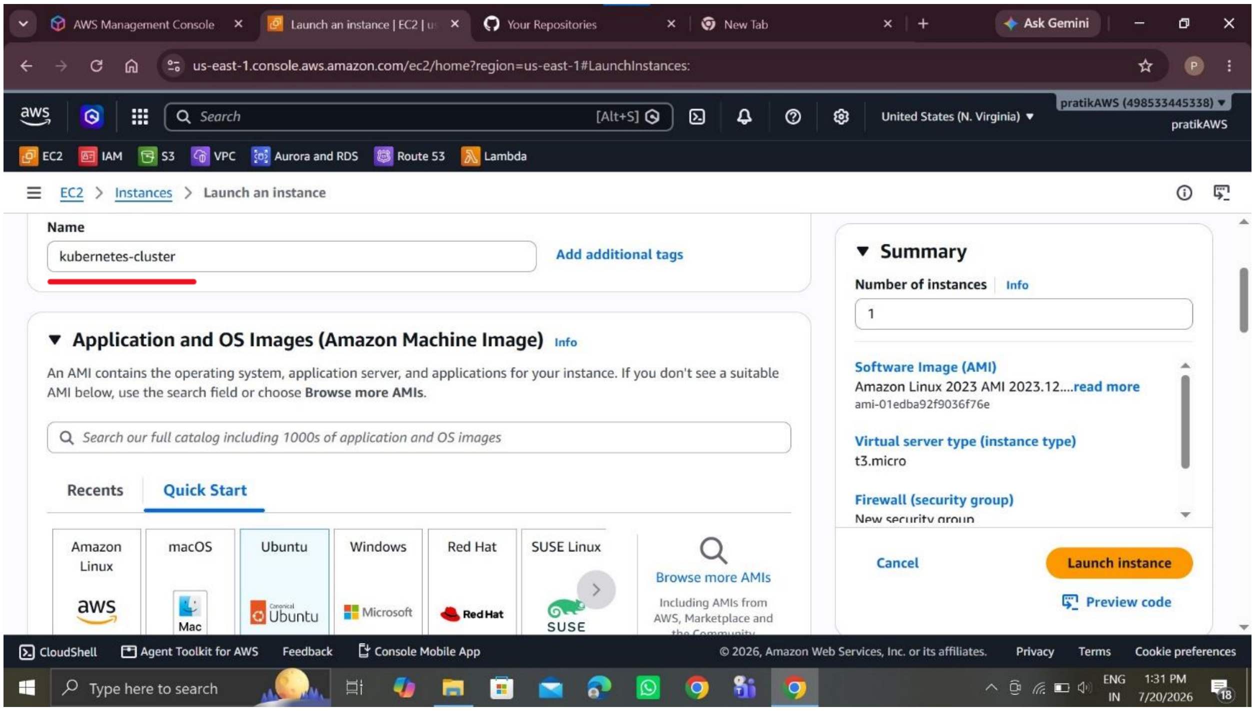
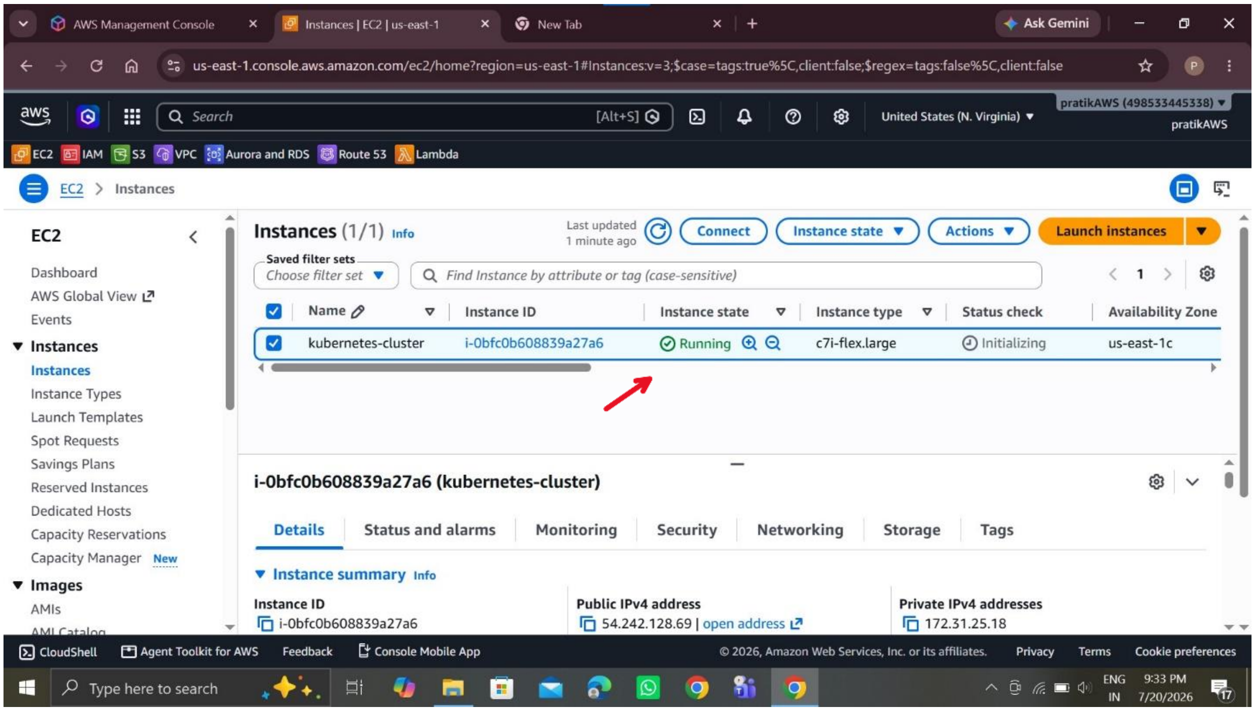
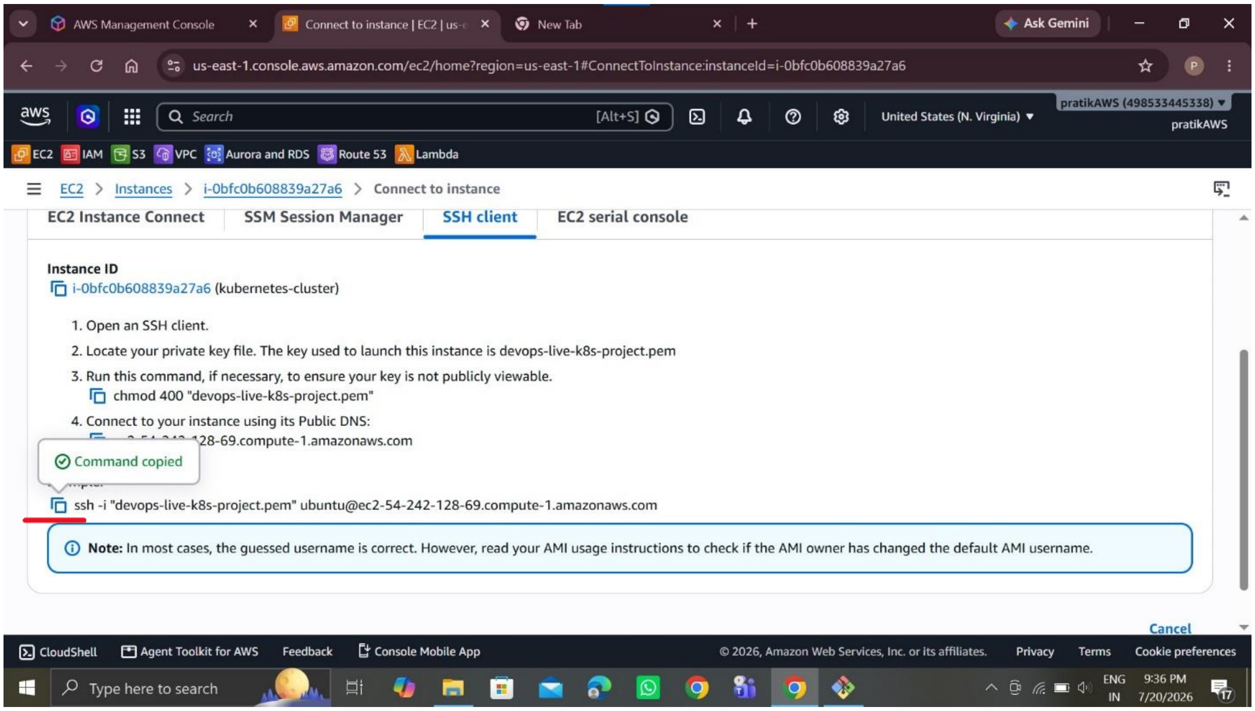
K8s Kind Voting App Deployment and configure argocd

Launch Kubernetes cluster

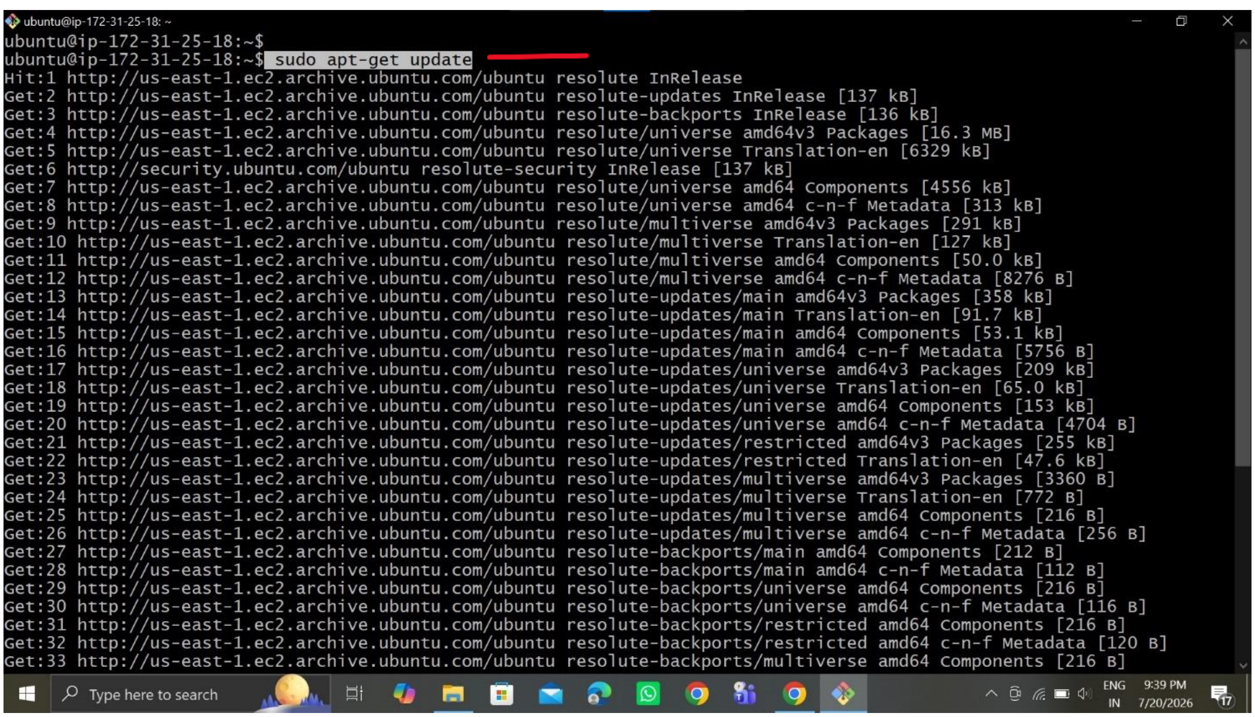




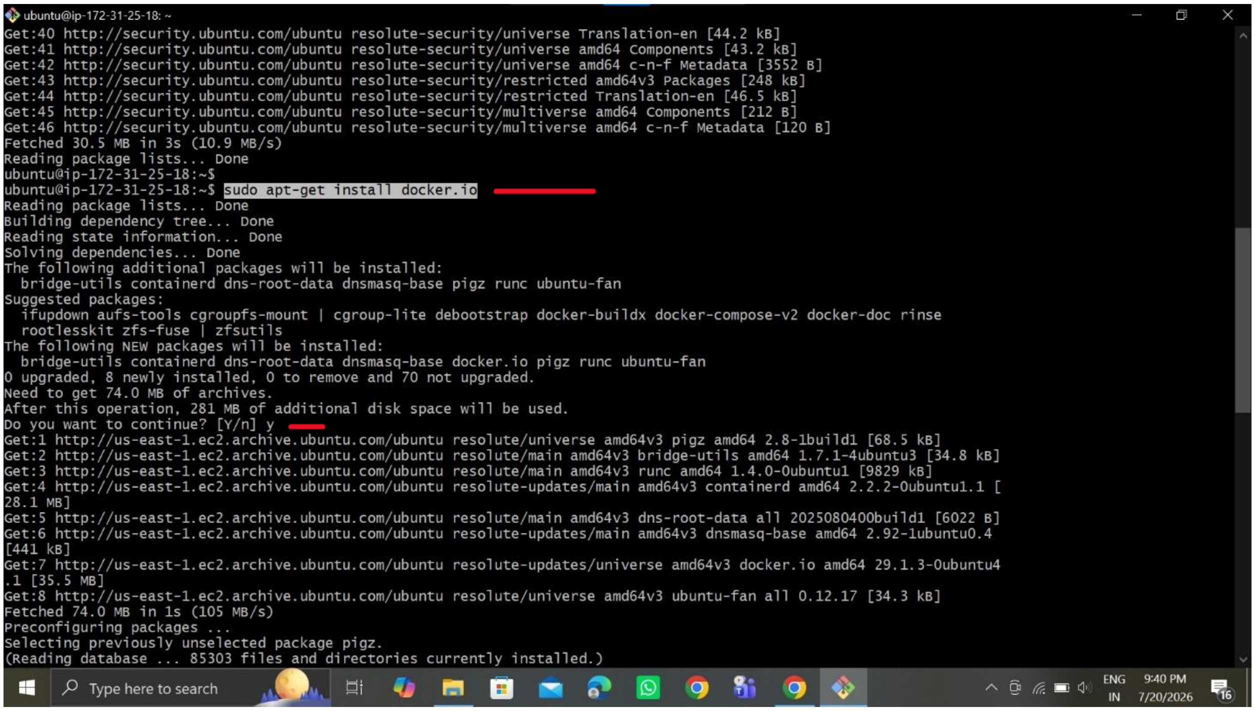
SSH through Login Kubernetes cluster in Git Bash Terminal



Update system

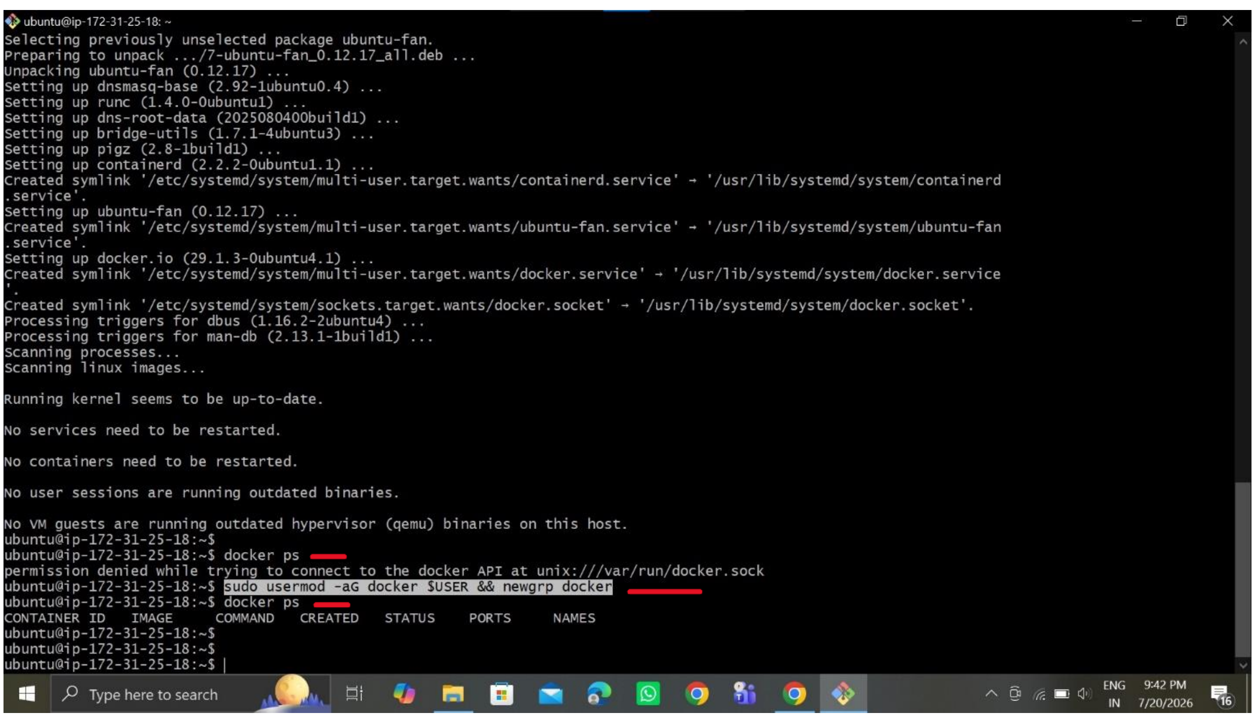


Install docker



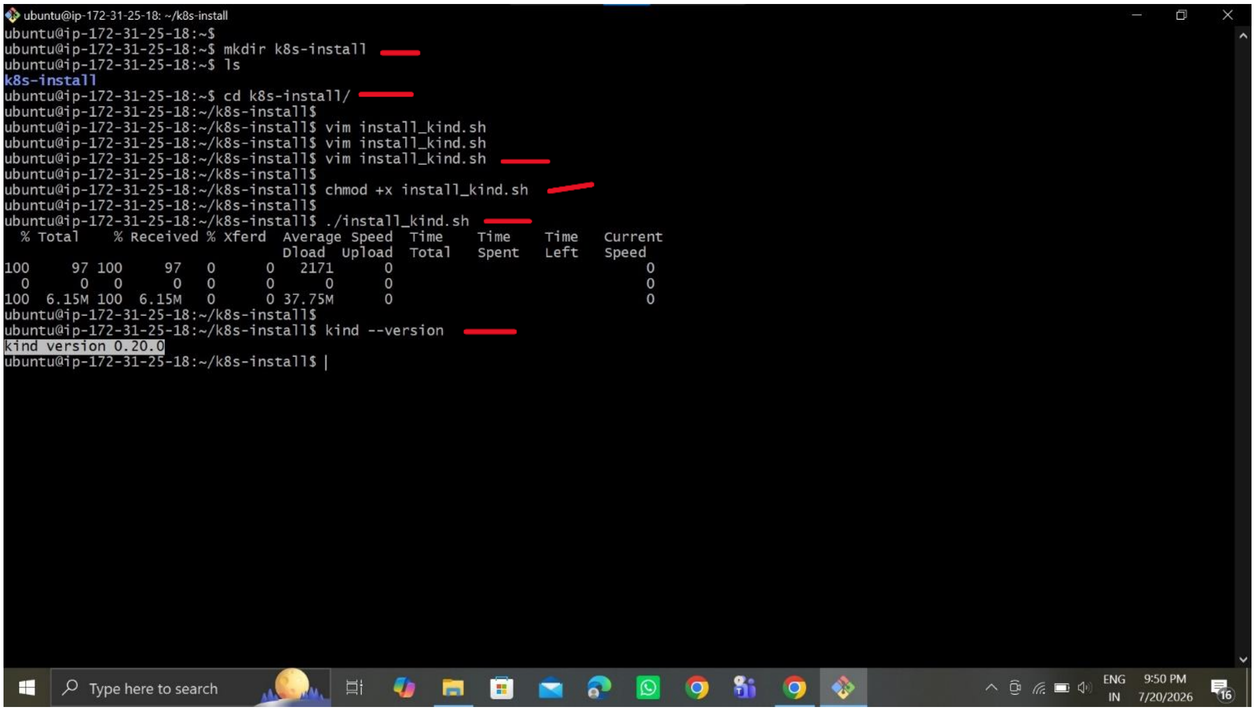
```
ubuntu@ip-172-31-25-18: ~
Get:40 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [44.2 kB]
Get:41 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:42 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:43 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64v3 Packages [248 kB]
Get:44 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [46.5 kB]
Get:45 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:46 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Fetched 30.5 MB in 3s (10.9 MB/s)
Reading package lists... Done
ubuntu@ip-172-31-25-18:~$
ubuntu@ip-172-31-25-18:~$ sudo apt-get install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse
  rootlesskit zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 70 not upgraded.
Need to get 74.0 MB of archives.
After this operation, 281 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 pigz amd64 2.8-1build1 [68.5 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 bridge-utils amd64 1.7.1-4ubuntu3 [34.8 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 runc amd64 1.4.0-0ubuntu1 [9829 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 containerd amd64 2.2.2-0ubuntu1.1 [
28.1 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 dns-root-data all 2025080400build1 [6022 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 dnsmasq-base amd64 2.92-1ubuntu0.4
[441 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 docker.io amd64 29.1.3-0ubuntu4
.1 [35.5 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/universe amd64v3 ubuntu-fan all 0.12.17 [34.3 kB]
Fetched 74.0 MB in 1s (105 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 85303 files and directories currently installed.)
```


Modify Docker USER permission

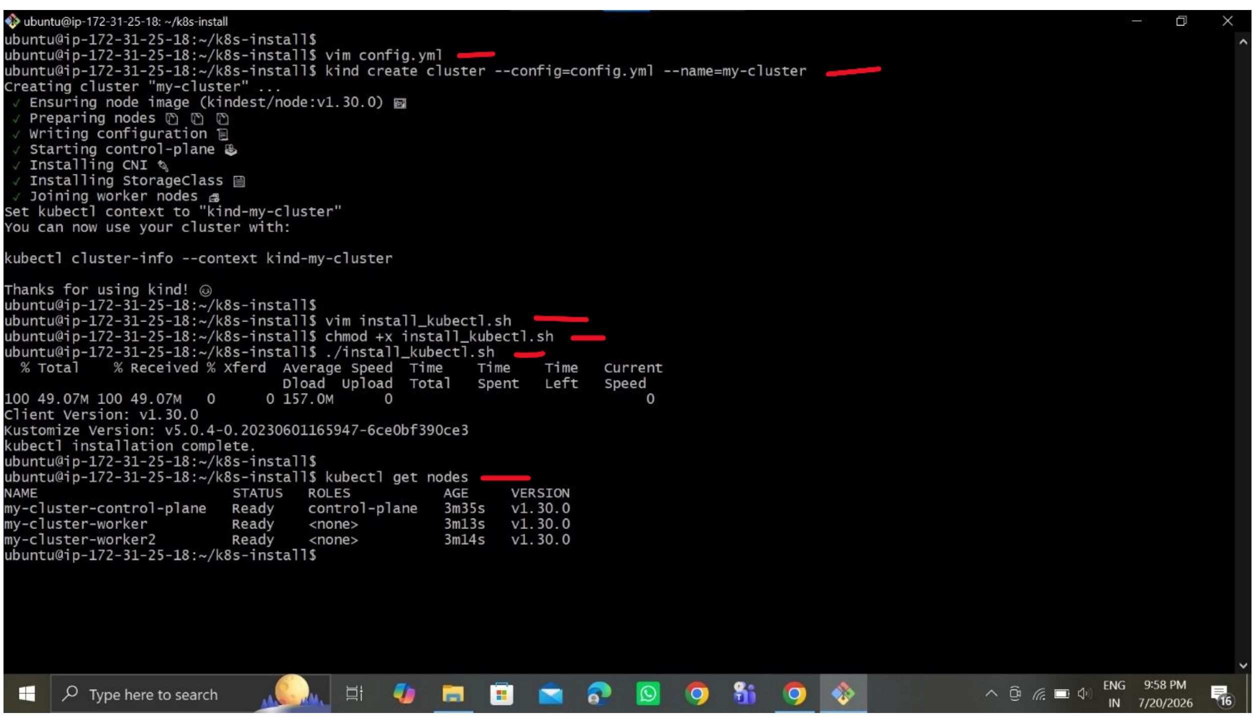


```
ubuntu@ip-172-31-25-18: ~
Selecting previously unselected package ubuntu-fan.
Preparing to unpack .../7-ubuntu-fan_0.12.17_all.deb ...
Unpacking ubuntu-fan (0.12.17) ...
Setting up dnsmasq-base (2.92-1ubuntu0.4) ...
Setting up runc (1.4.0-0ubuntu1) ...
Setting up dns-root-data (2025080400build1) ...
Setting up bridge-utils (1.7.1-4ubuntu3) ...
Setting up pigz (2.8-1build1) ...
Setting up containerd (2.2.2-0ubuntu1.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' → '/usr/lib/systemd/system/containerd
.service'.
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' → '/usr/lib/systemd/system/ubuntu-fan
.service'.
Setting up docker.io (29.1.3-0ubuntu4.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service
'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/usr/lib/systemd/system/docker.socket'.
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-25-18:~$
ubuntu@ip-172-31-25-18:~$ docker ps
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-25-18:~$ sudo usermod -aG docker $USER && newgrp docker
ubuntu@ip-172-31-25-18:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-25-18:~$
ubuntu@ip-172-31-25-18:~$
ubuntu@ip-172-31-25-18:~$ |
```

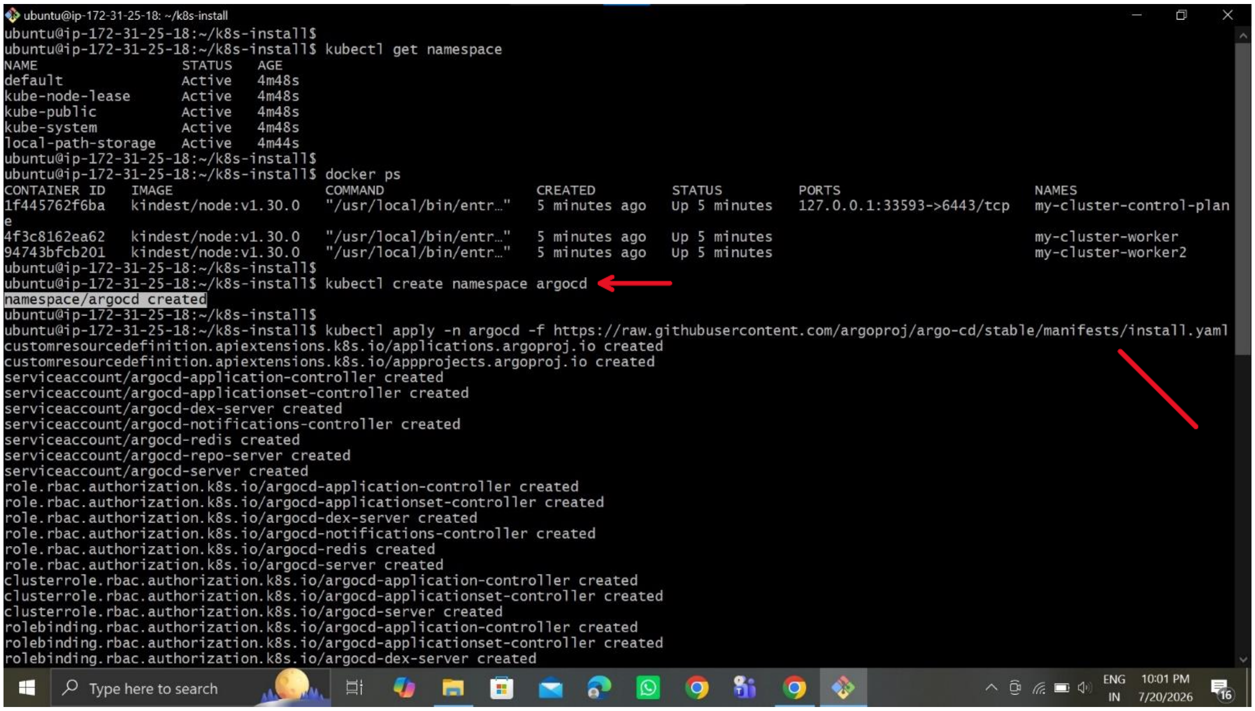
Install kind



Configure kind cluster & install kubectl

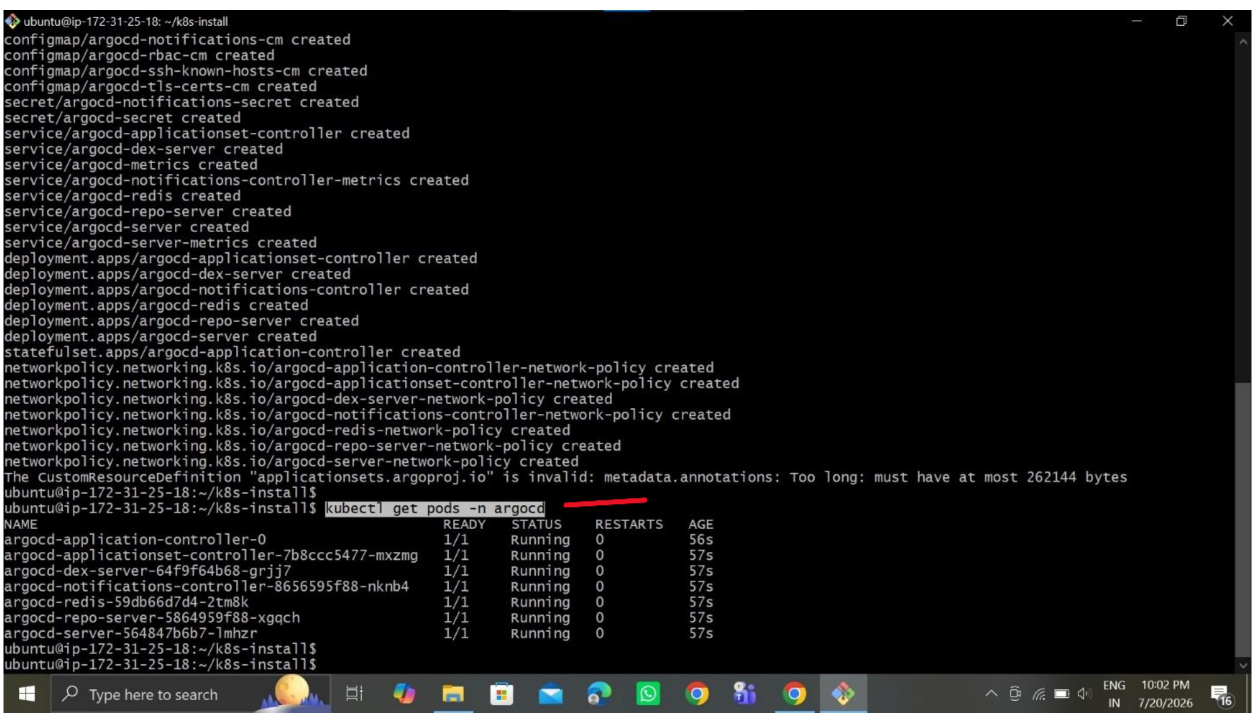


Create argocd namespace



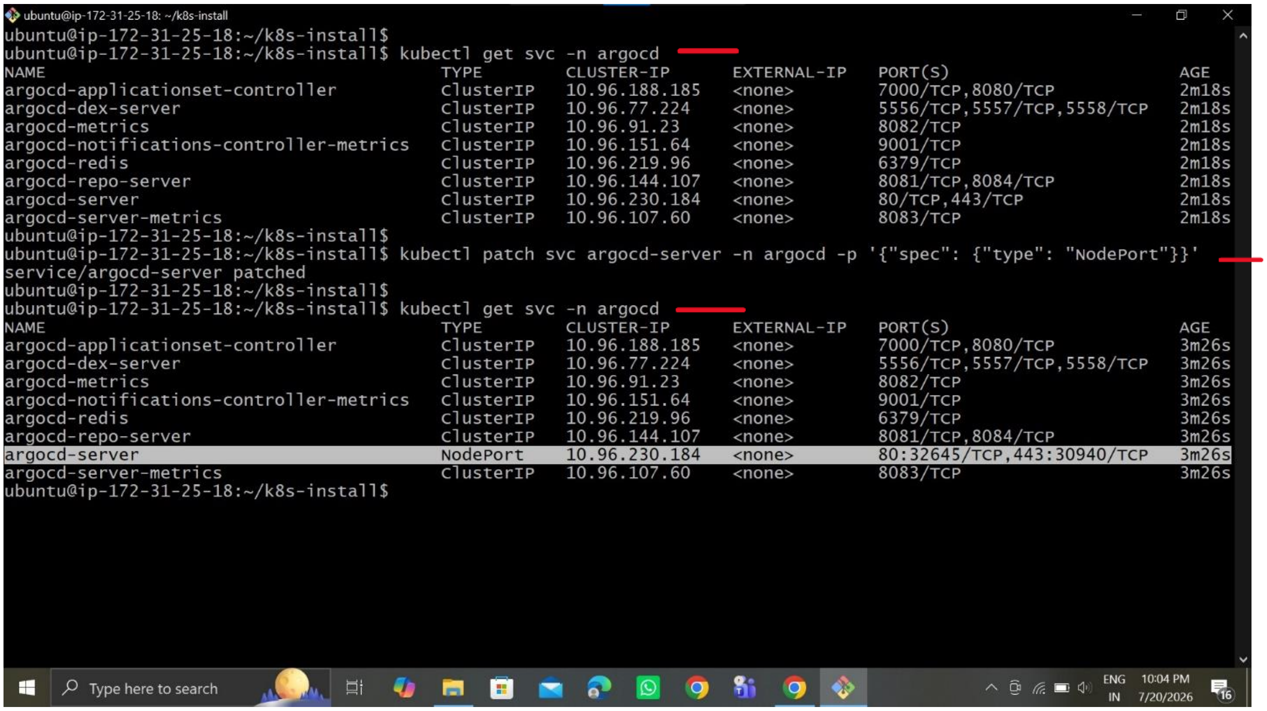
```
ubuntu@ip-172-31-25-18: ~/k8s-install
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get namespace
NAME                 STATUS   AGE
default              Active   4m48s
kube-node-lease      Active   4m48s
kube-public          Active   4m48s
kube-system          Active   4m48s
local-path-storage   Active   4m44s
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED         STATUS         PORTS                       NAMES
1f445762f6ba   kindest/node:v1.30.0   "/usr/local/bin/entr…"   5 minutes ago   Up 5 minutes   127.0.0.1:33593->6443/tcp   my-cluster-control-plan
e
4f3c8162ea62   kindest/node:v1.30.0   "/usr/local/bin/entr…"   5 minutes ago   Up 5 minutes                               my-cluster-worker
94743bfcb201   kindest/node:v1.30.0   "/usr/local/bin/entr…"   5 minutes ago   Up 5 minutes                               my-cluster-worker2
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl create namespace argocd
namespace/argocd created
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl apply -n argocd -f https://raw.githubusercontent.com/argoproj/argo-cd/stable/manifests/install.yaml
customresourcedefinition.apiextensions.k8s.io/applications.argoproj.io created
customresourcedefinition.apiextensions.k8s.io/appprojects.argoproj.io created
serviceaccount/argocd-application-controller created
serviceaccount/argocd-applicationset-controller created
serviceaccount/argocd-dex-server created
serviceaccount/argocd-notifications-controller created
serviceaccount/argocd-redis created
serviceaccount/argocd-repo-server created
serviceaccount/argocd-server created
role.rbac.authorization.k8s.io/argocd-application-controller created
role.rbac.authorization.k8s.io/argocd-applicationset-controller created
role.rbac.authorization.k8s.io/argocd-dex-server created
role.rbac.authorization.k8s.io/argocd-notifications-controller created
role.rbac.authorization.k8s.io/argocd-redis created
role.rbac.authorization.k8s.io/argocd-server created
clusterrole.rbac.authorization.k8s.io/argocd-application-controller created
clusterrole.rbac.authorization.k8s.io/argocd-applicationset-controller created
clusterrole.rbac.authorization.k8s.io/argocd-server created
rolebinding.rbac.authorization.k8s.io/argocd-application-controller created
rolebinding.rbac.authorization.k8s.io/argocd-applicationset-controller created
rolebinding.rbac.authorization.k8s.io/argocd-dex-server created
```


Successfully show argocd pods

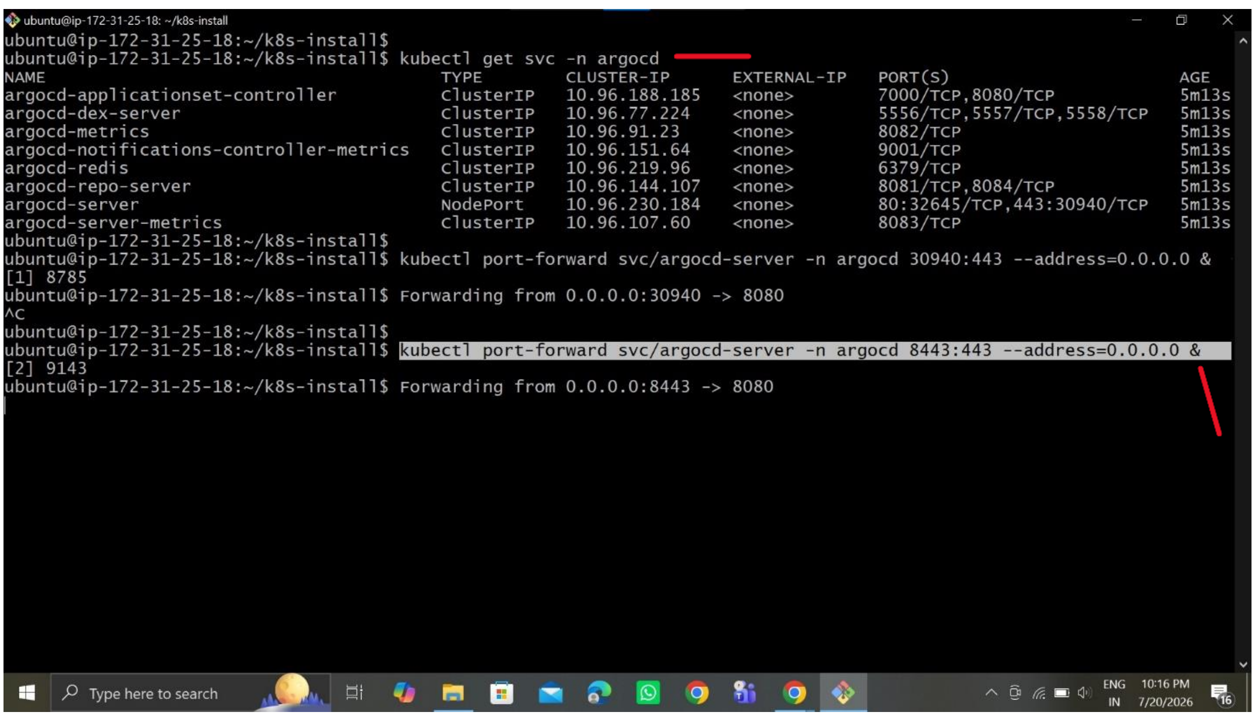


```
ubuntu@ip-172-31-25-18: ~/k8s-install
configmap/argocd-notifications-cm created
configmap/argocd-rbac-cm created
configmap/argocd-ssh-known-hosts-cm created
configmap/argocd-tls-certs-cm created
secret/argocd-notifications-secret created
secret/argocd-secret created
service/argocd-applicationset-controller created
service/argocd-dex-server created
service/argocd-metrics created
service/argocd-notifications-controller-metrics created
service/argocd-redis created
service/argocd-repo-server created
service/argocd-server created
service/argocd-server-metrics created
deployment.apps/argocd-applicationset-controller created
deployment.apps/argocd-dex-server created
deployment.apps/argocd-notifications-controller created
deployment.apps/argocd-redis created
deployment.apps/argocd-repo-server created
deployment.apps/argocd-server created
statefulset.apps/argocd-application-controller created
networkpolicy.networking.k8s.io/argocd-application-controller-network-policy created
networkpolicy.networking.k8s.io/argocd-applicationset-controller-network-policy created
networkpolicy.networking.k8s.io/argocd-dex-server-network-policy created
networkpolicy.networking.k8s.io/argocd-notifications-controller-network-policy created
networkpolicy.networking.k8s.io/argocd-redis-network-policy created
networkpolicy.networking.k8s.io/argocd-repo-server-network-policy created
networkpolicy.networking.k8s.io/argocd-server-network-policy created
The CustomResourceDefinition "applicationsets.argoproj.io" is invalid: metadata.annotations: Too long: must have at most 262144 bytes
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods -n argocd
NAME                                                READY   STATUS    RESTARTS   AGE
argocd-application-controller-0                     1/1     Running   0          56s
argocd-applicationset-controller-7b8ccc5477-mxzmg   1/1     Running   0          57s
argocd-dex-server-64f9f64b68-grjj7                  1/1     Running   0          57s
argocd-notifications-controller-8656595f88-nknb4    1/1     Running   0          57s
argocd-redis-59db66d7d4-2tm8k                       1/1     Running   0          57s
argocd-repo-server-5864959f88-xgqch                 1/1     Running   0          57s
argocd-server-564847b6b7-lmhzr                      1/1     Running   0          57s
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$
```

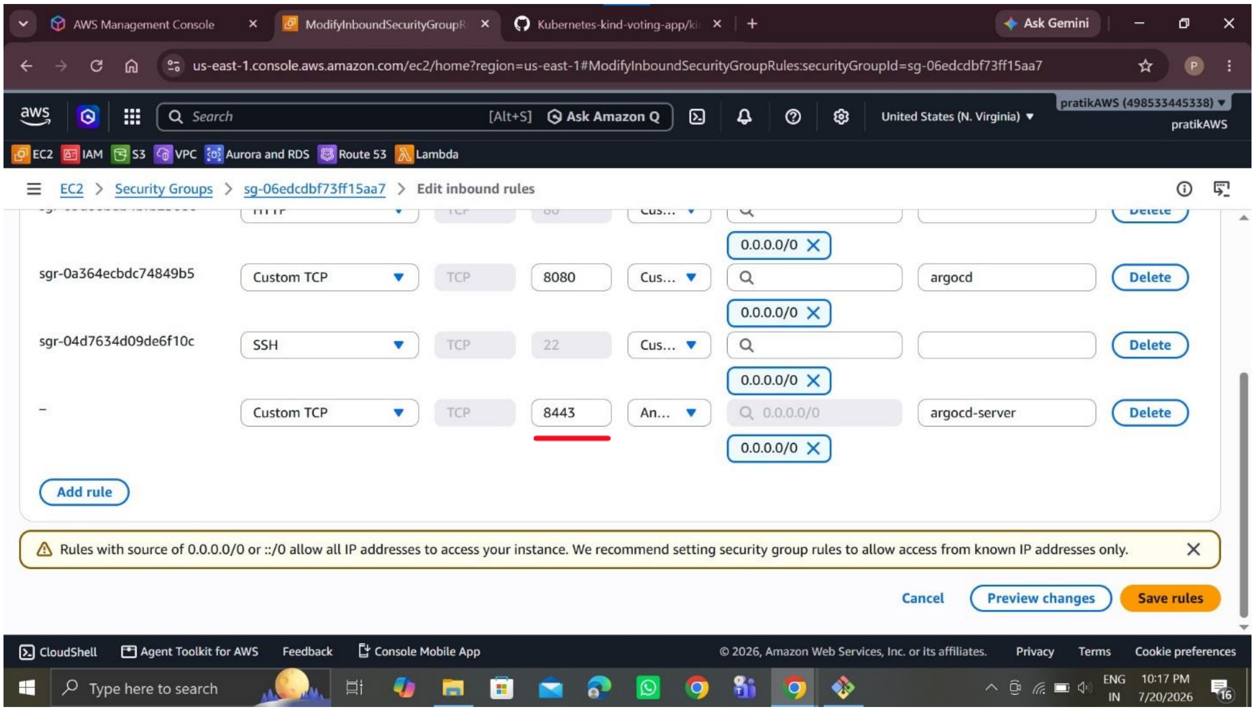
Then patch argocd server in Node port



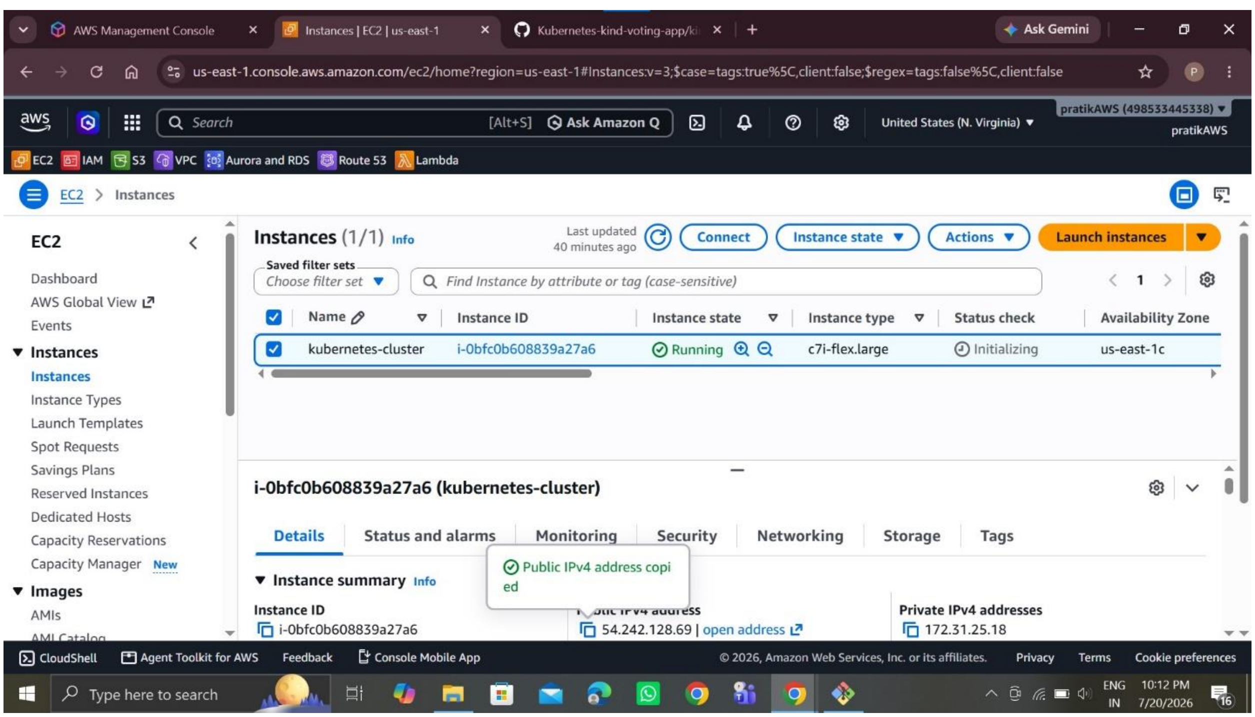
Port forward in argocd server



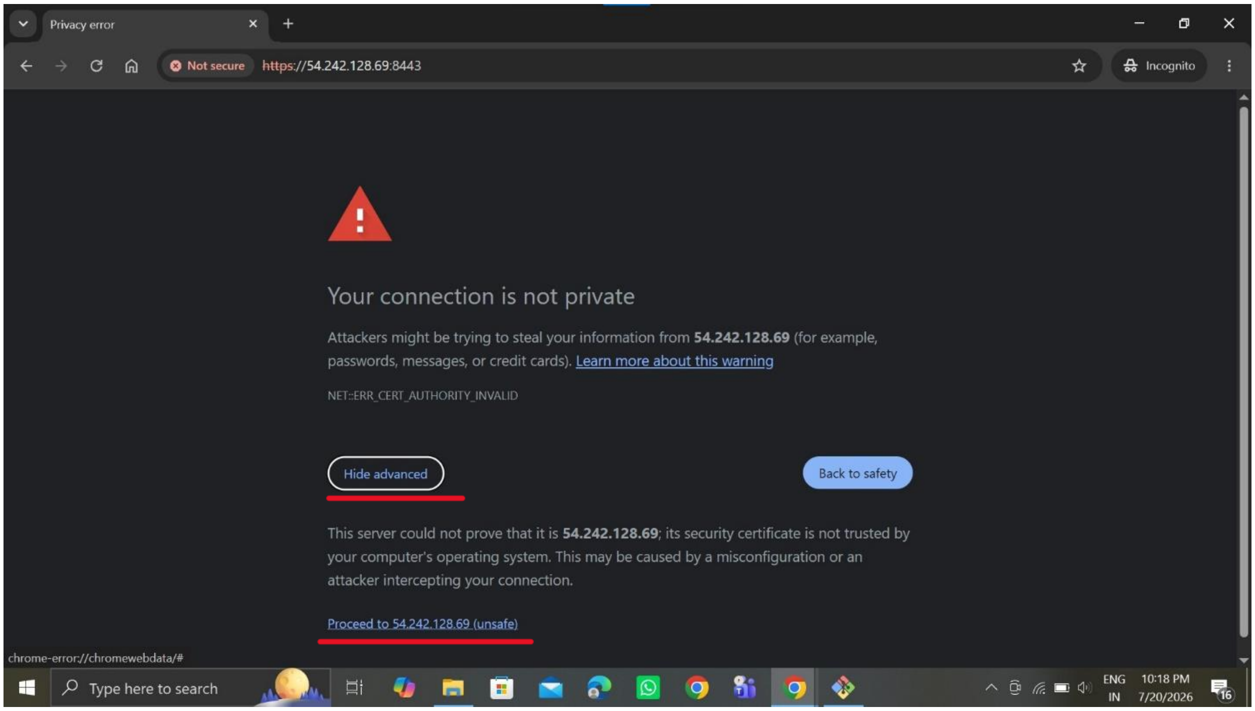
Allow this port in Security group



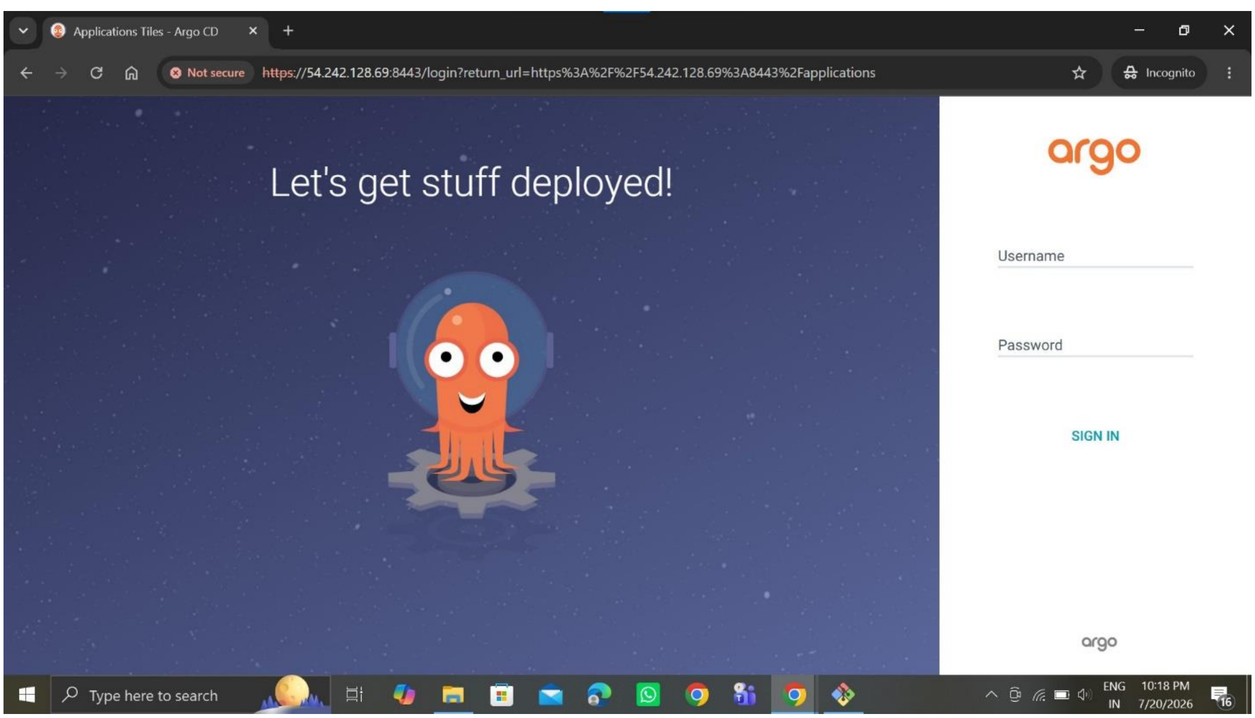
Then copy Public IP of Kubernetes cluster & go to browser



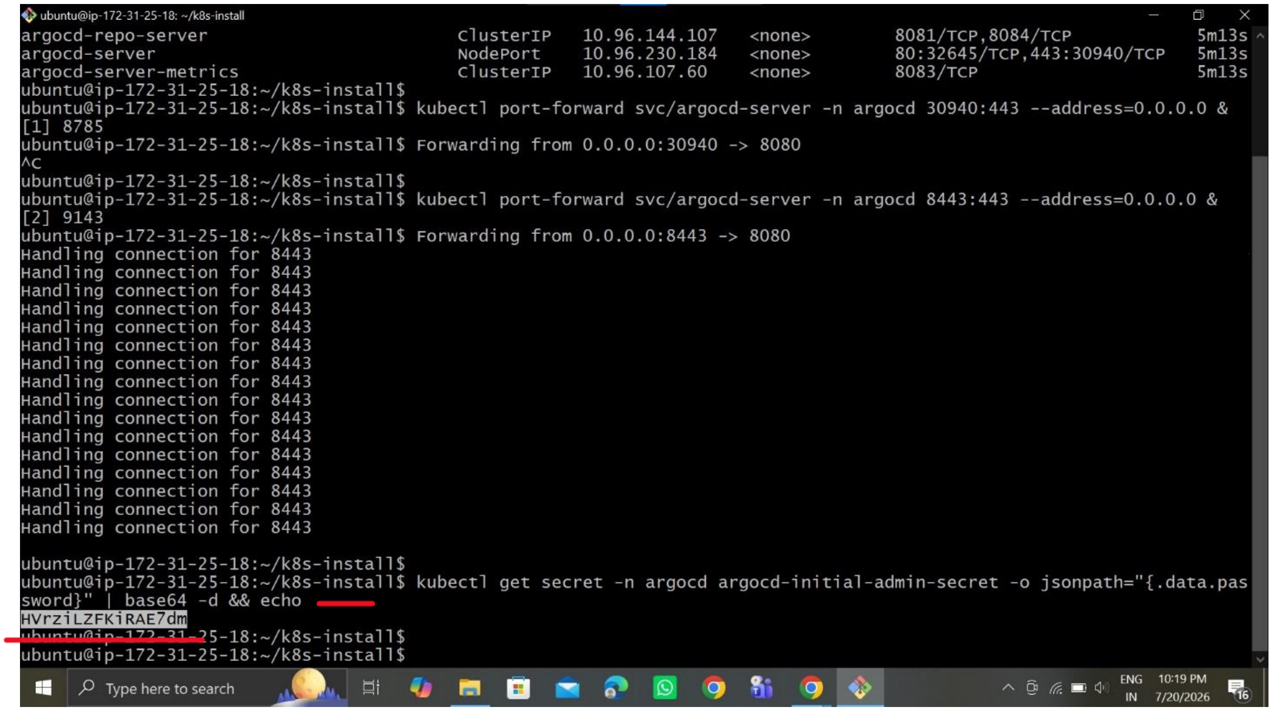
Paste this public IP & click Proceed



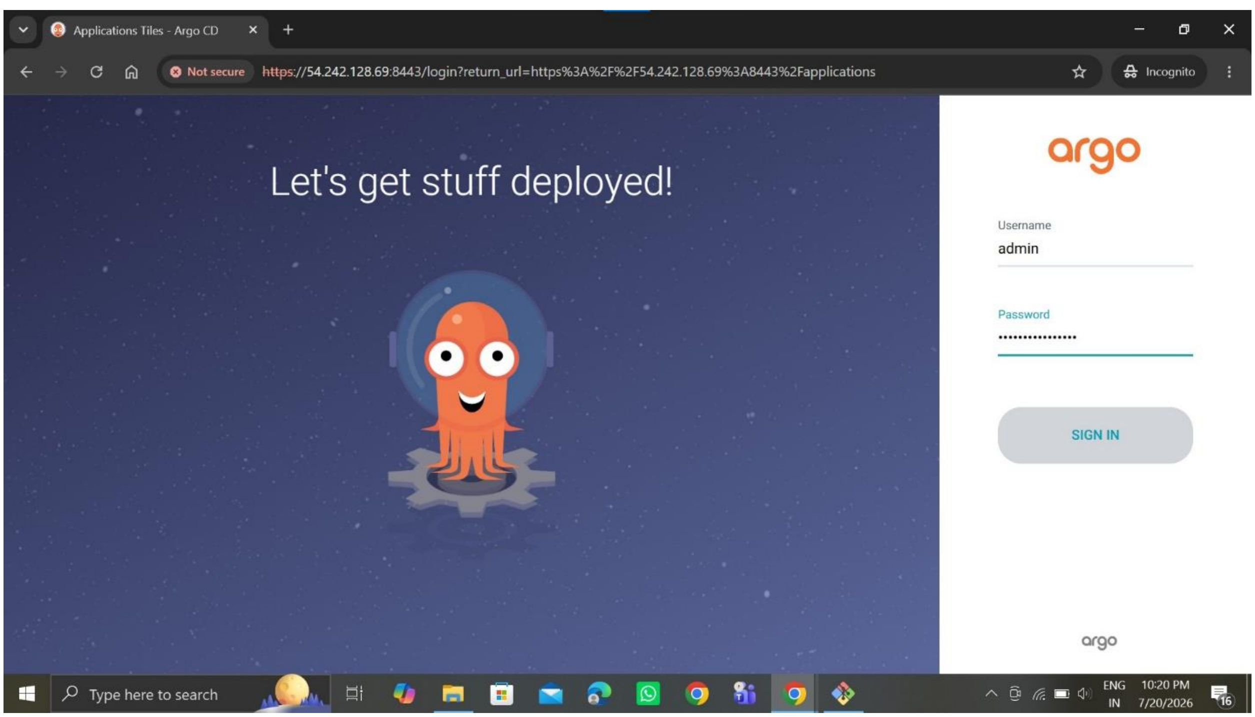
Successfully show the argocd login page



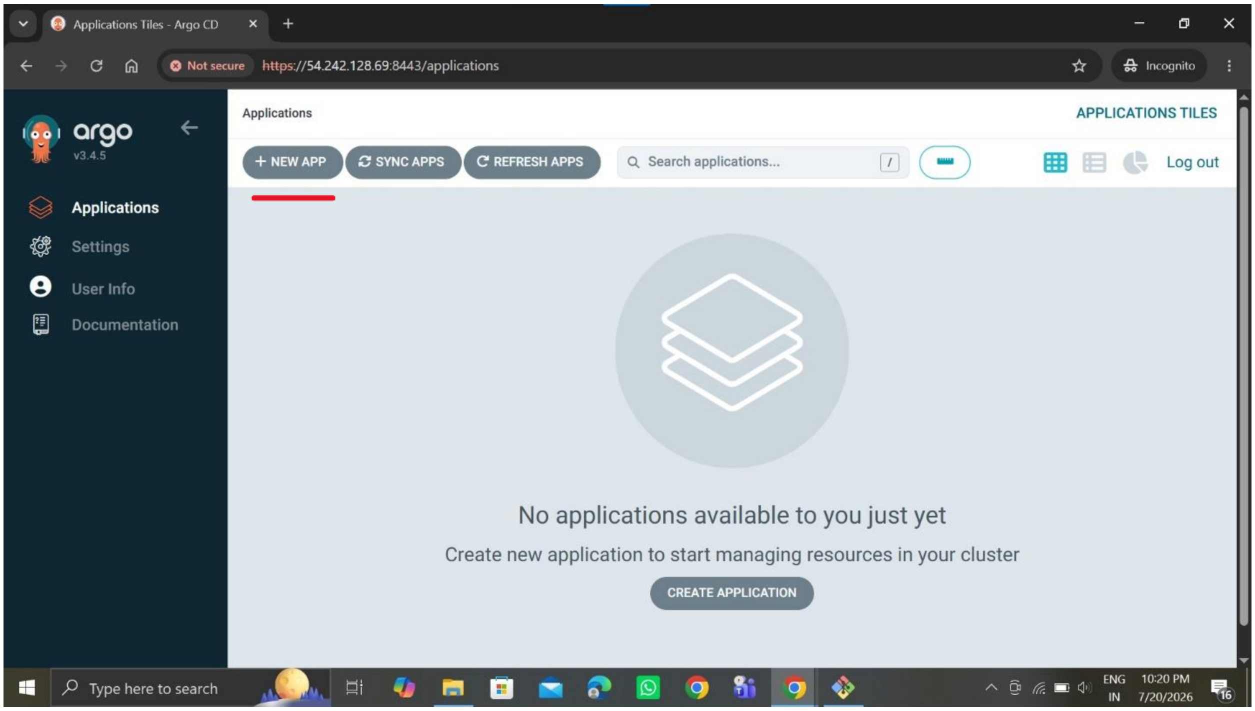
Then generate argocd initial admin password



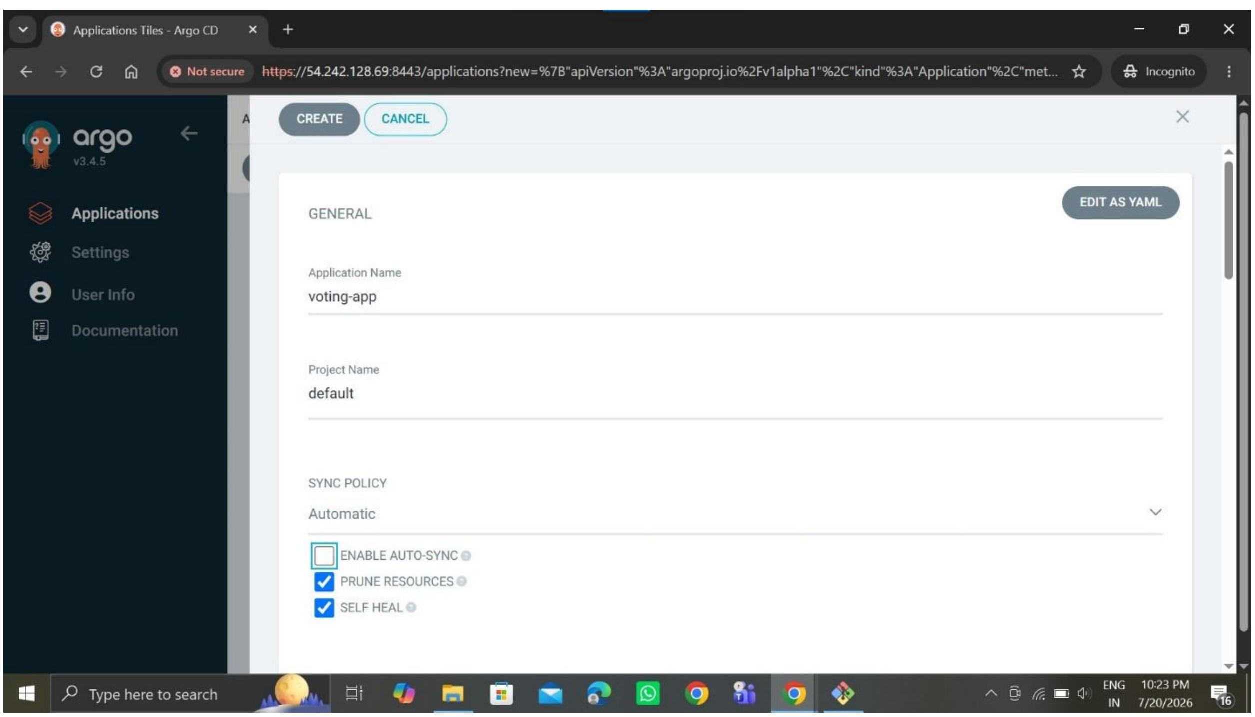
Then give the user name , password & then sign in



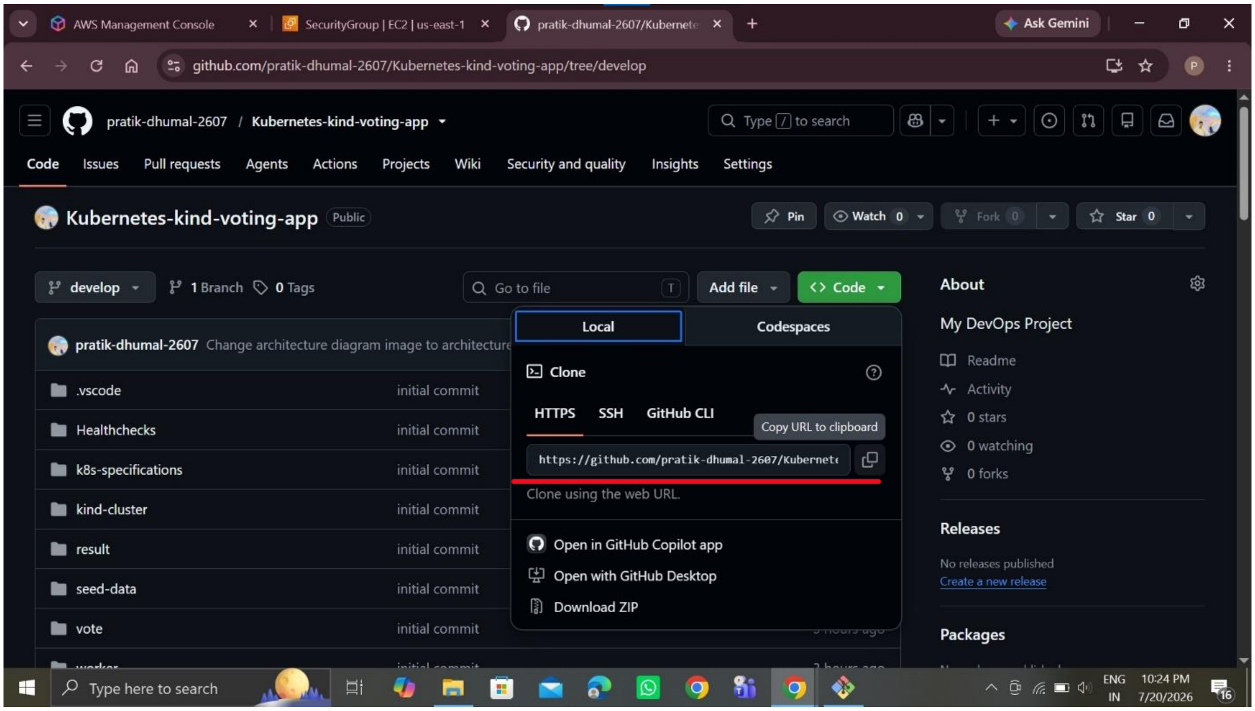
Successfully show argocd dashboard then click New App



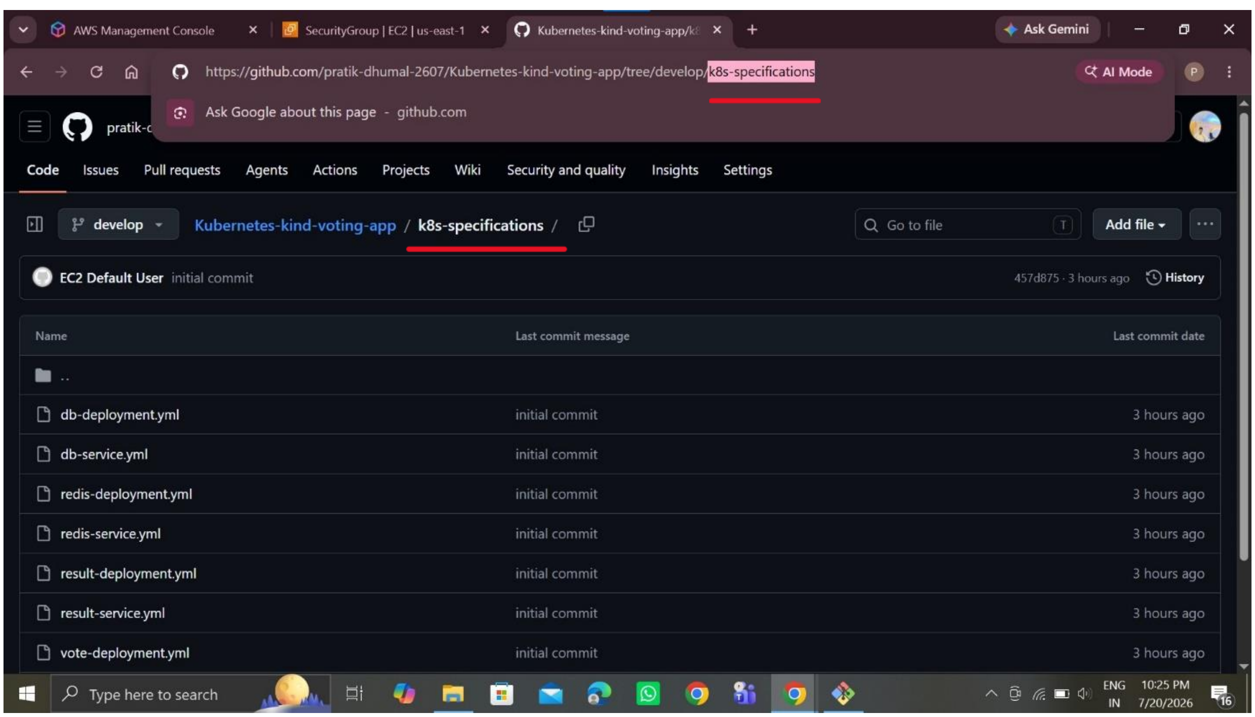
Fill application info



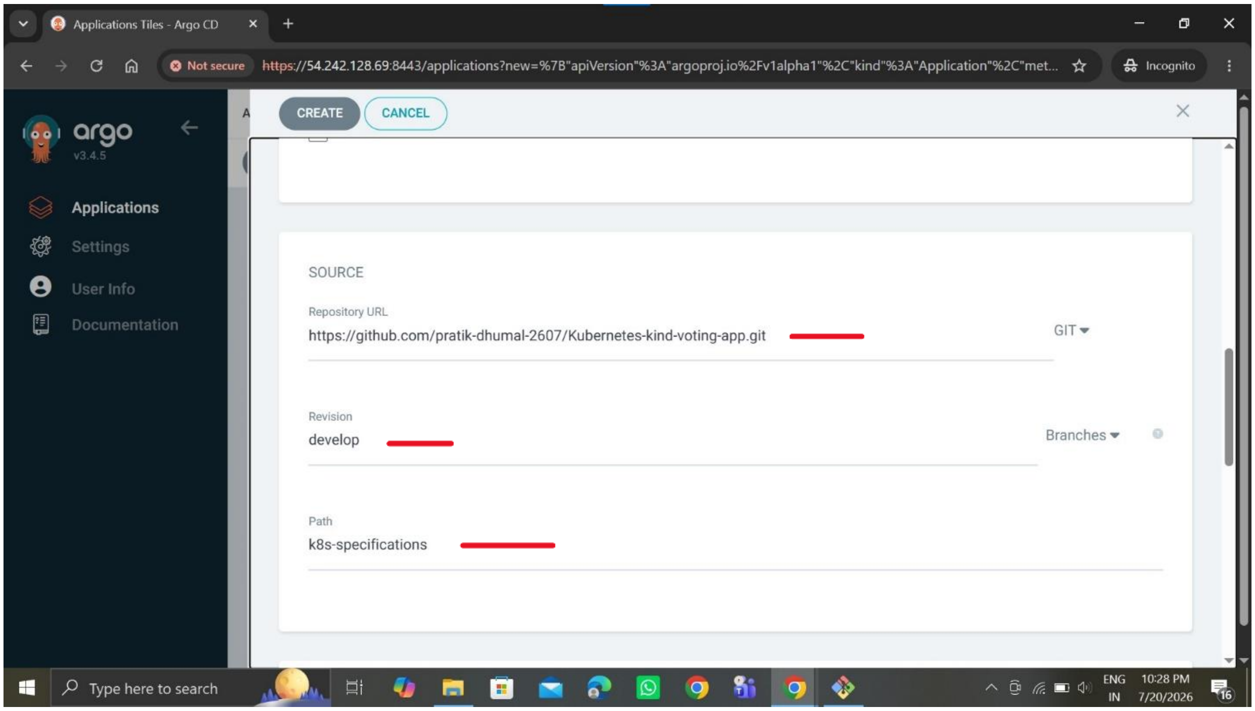
Copy github repository URL



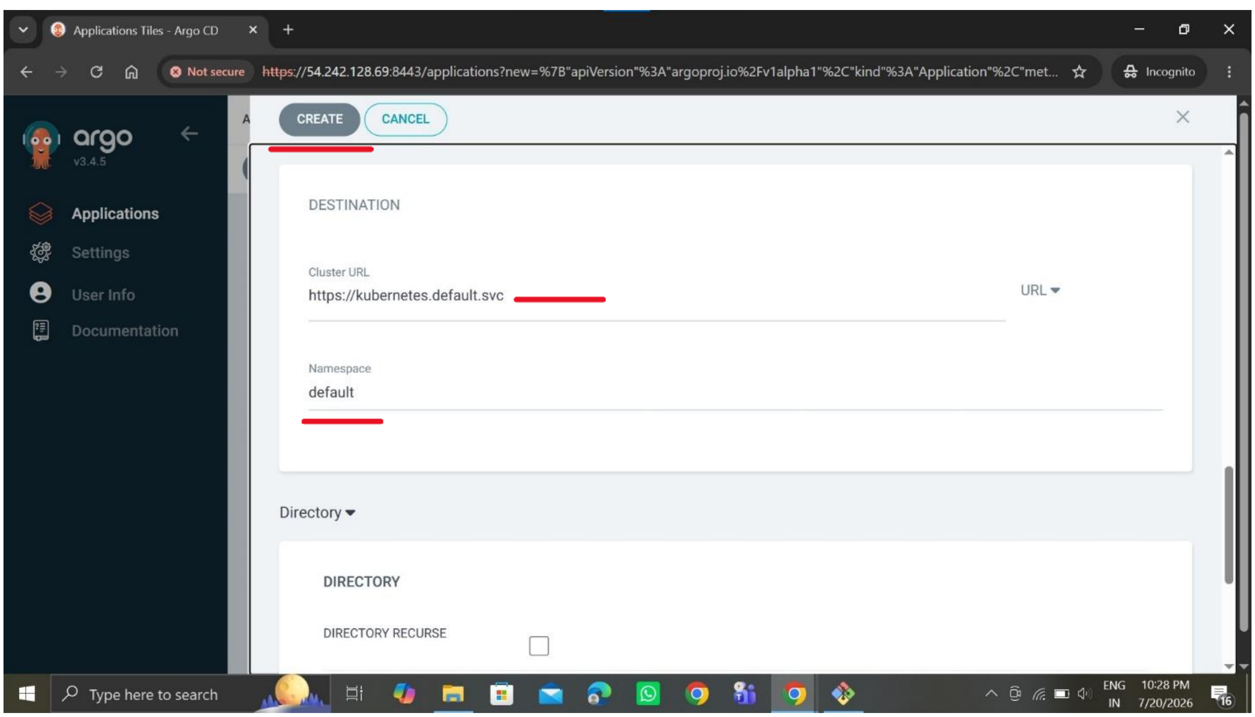
Copy the Path of manifest files folder



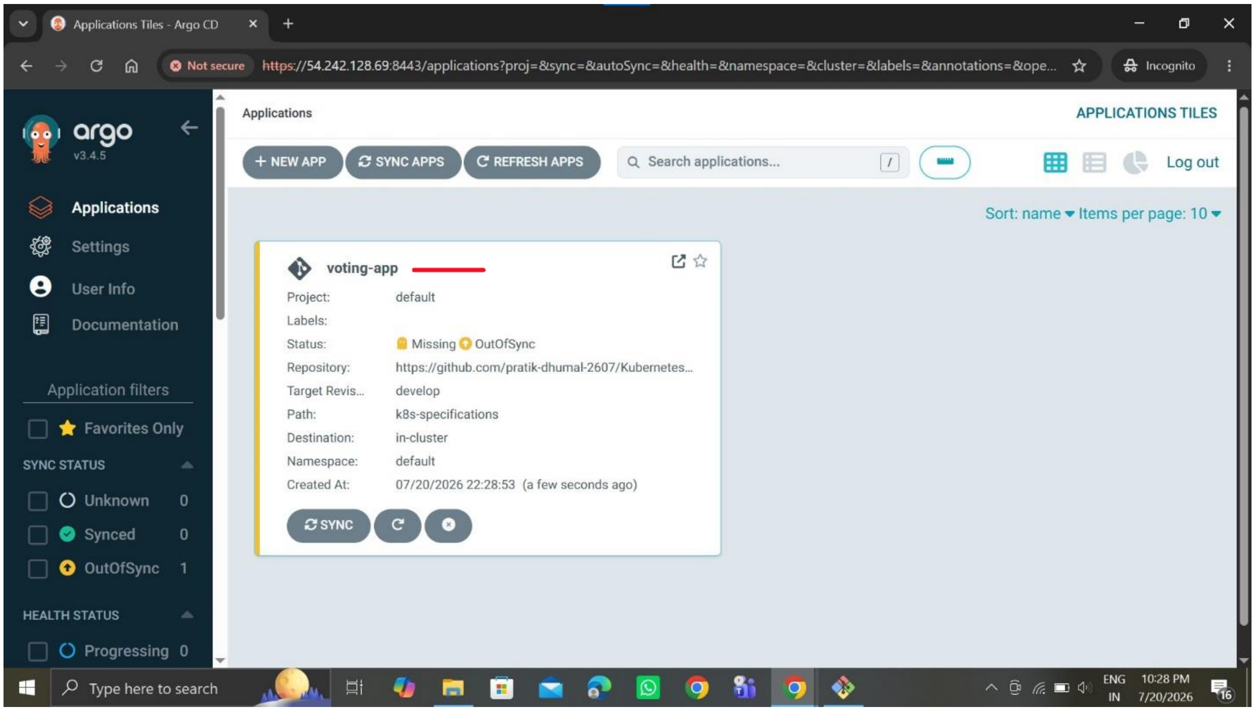
Paste this repository URL , select repository branch , paste path of manifest files folder



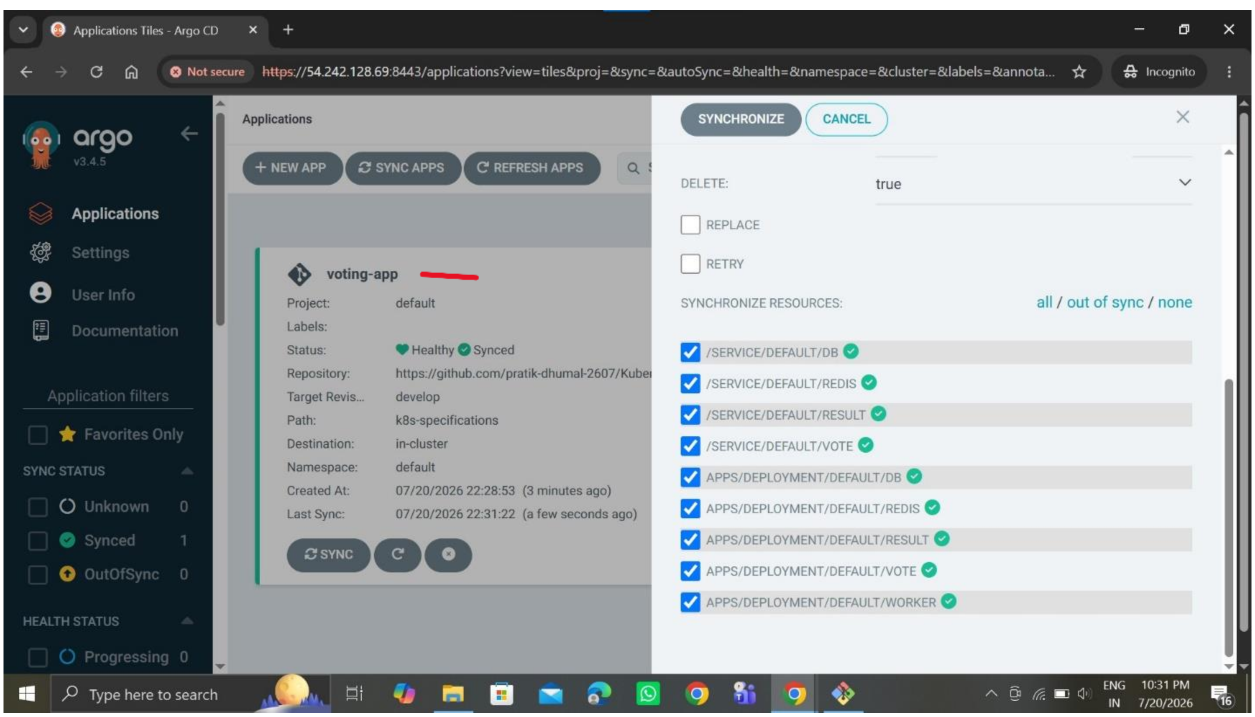
Then set cluster URL & set namespace default then click Create



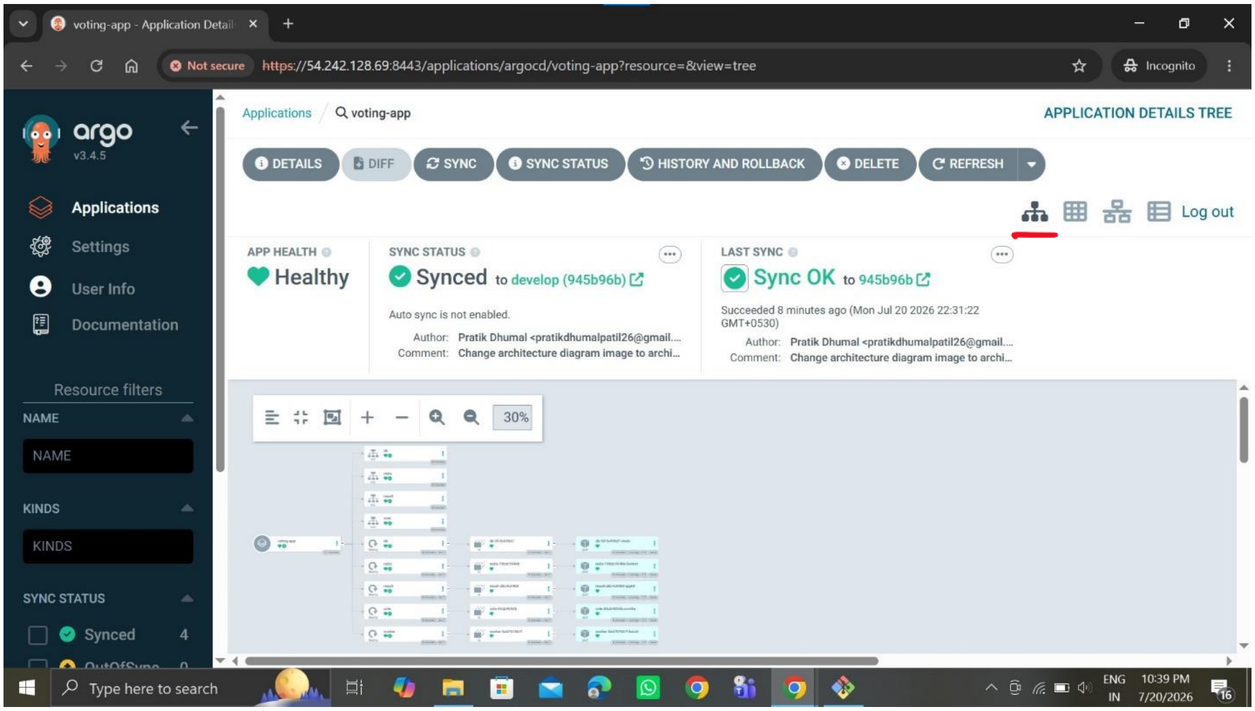
Application successfully created



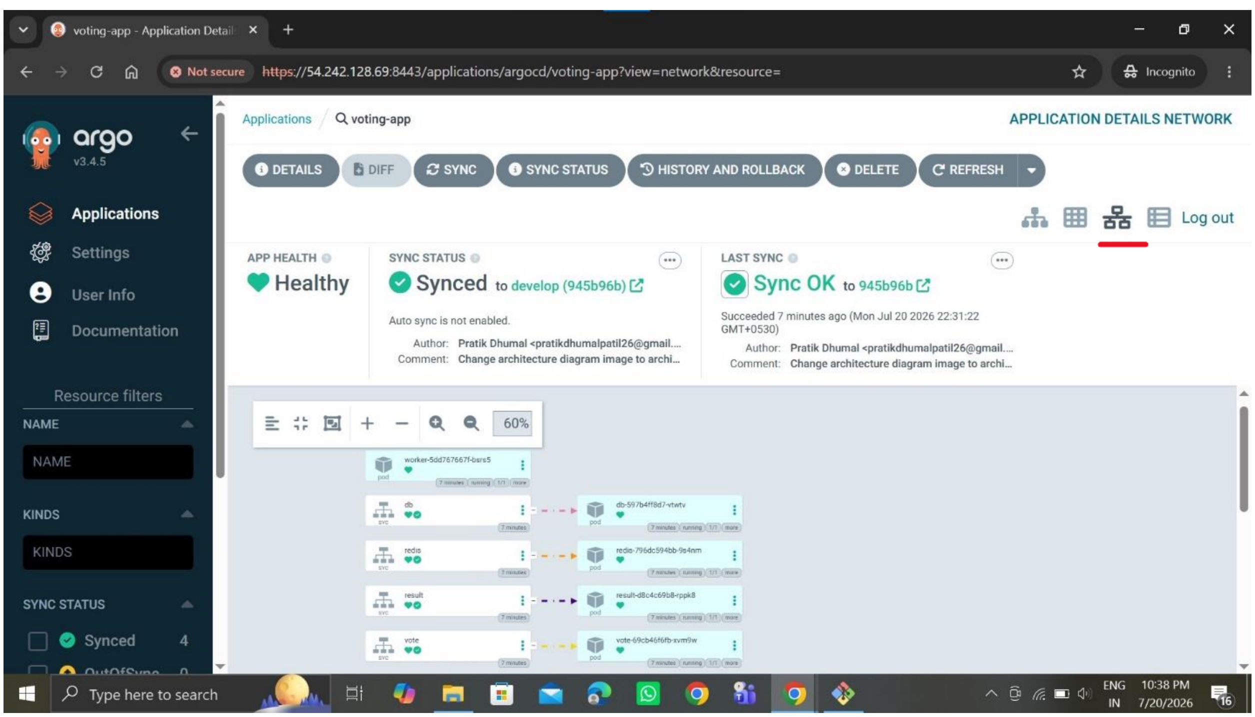
Then click this application



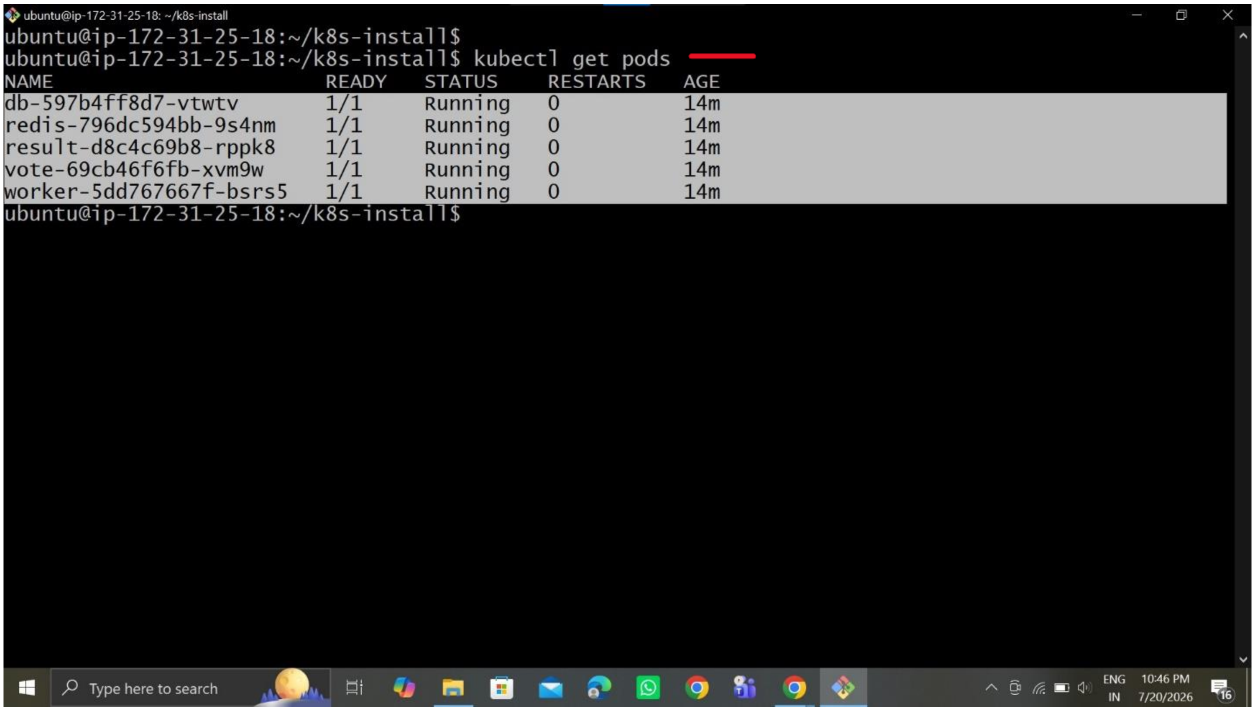
Successfully run this application



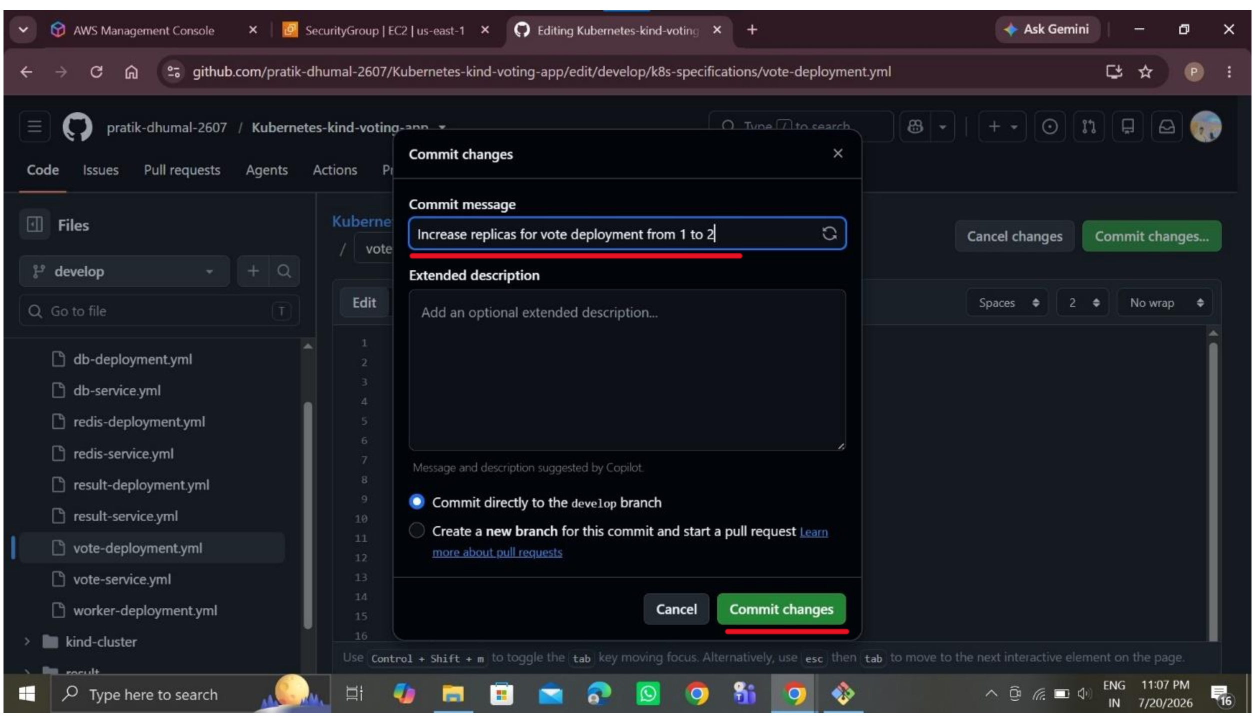
Then show this application in another format



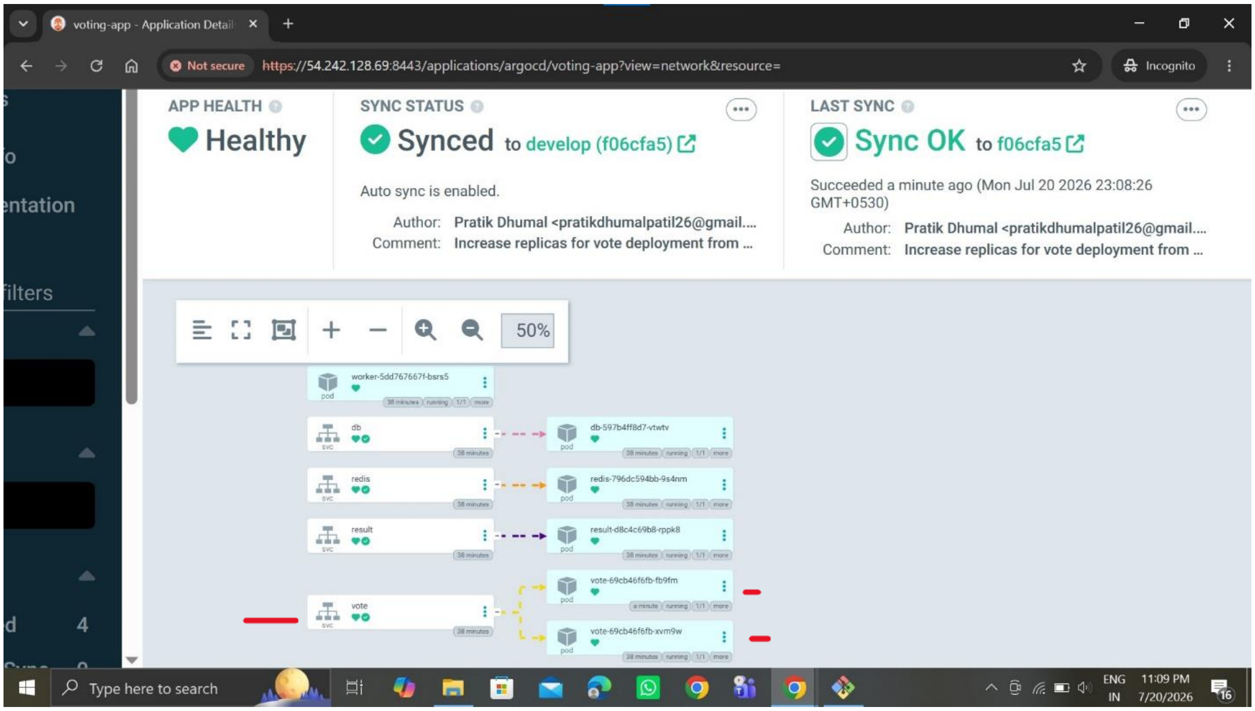
Then this pods show in terminal



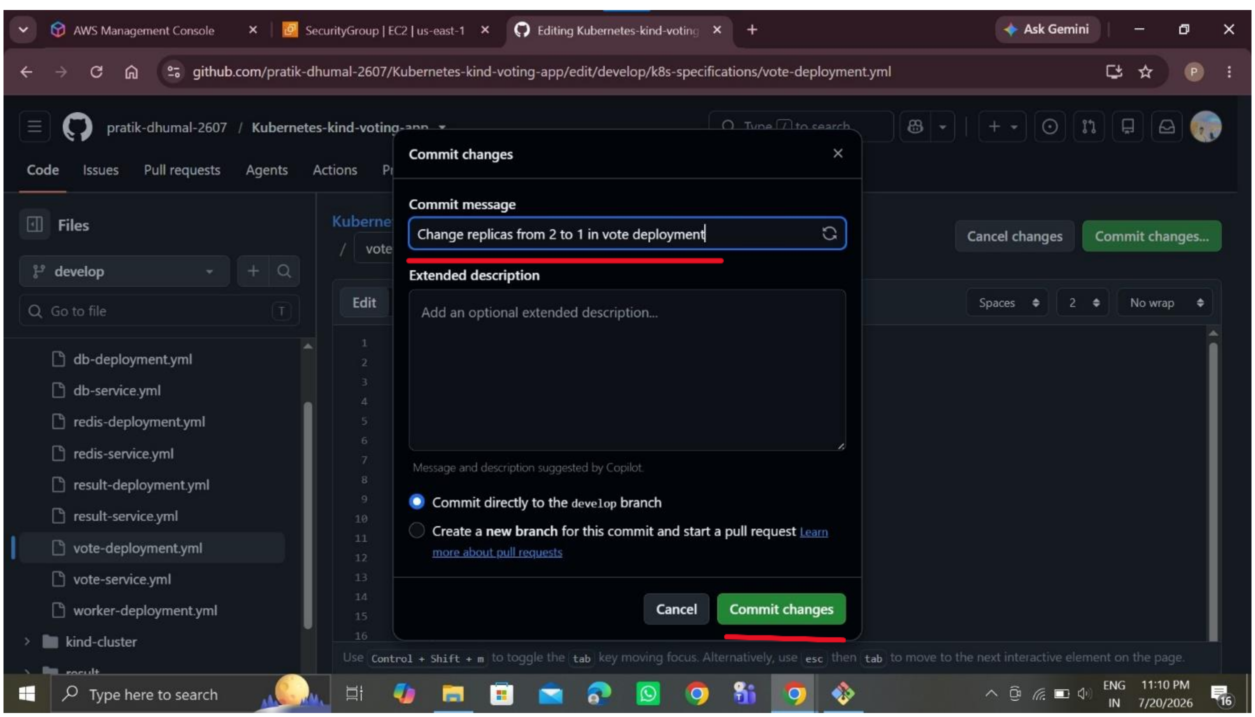
Then go to github & set vote replicas 1 to 2



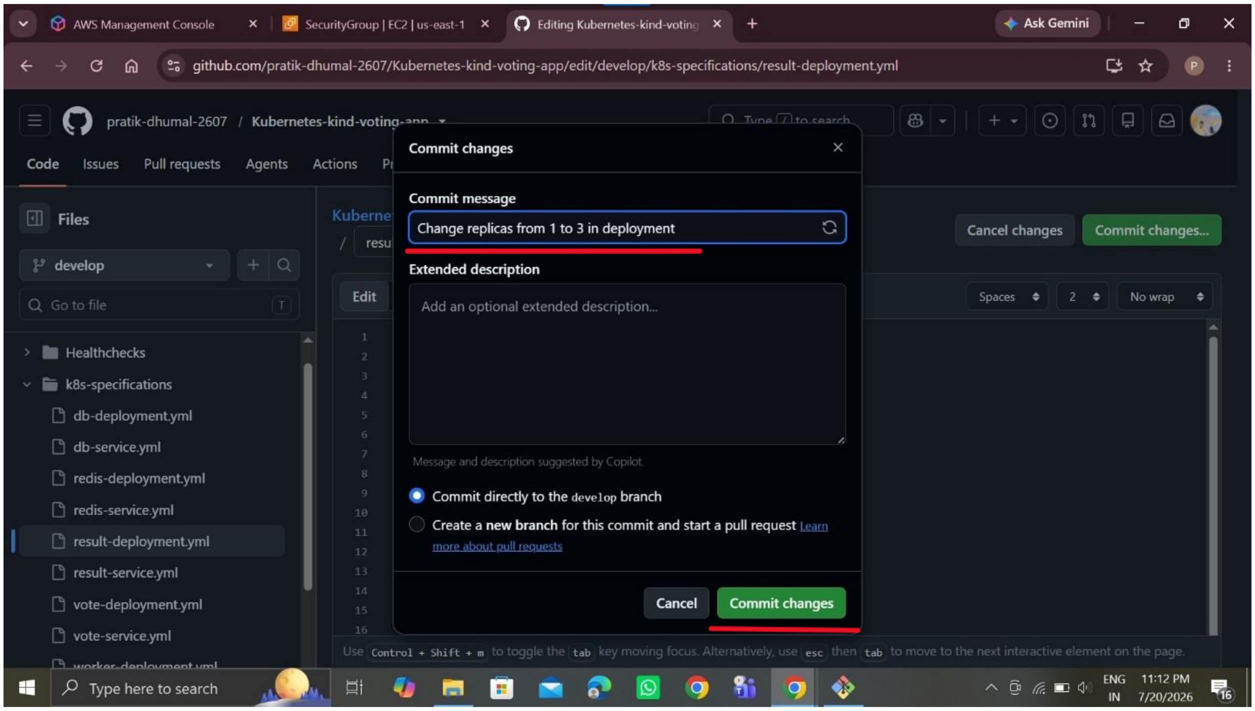
Successfully running 2 pods vote deployment



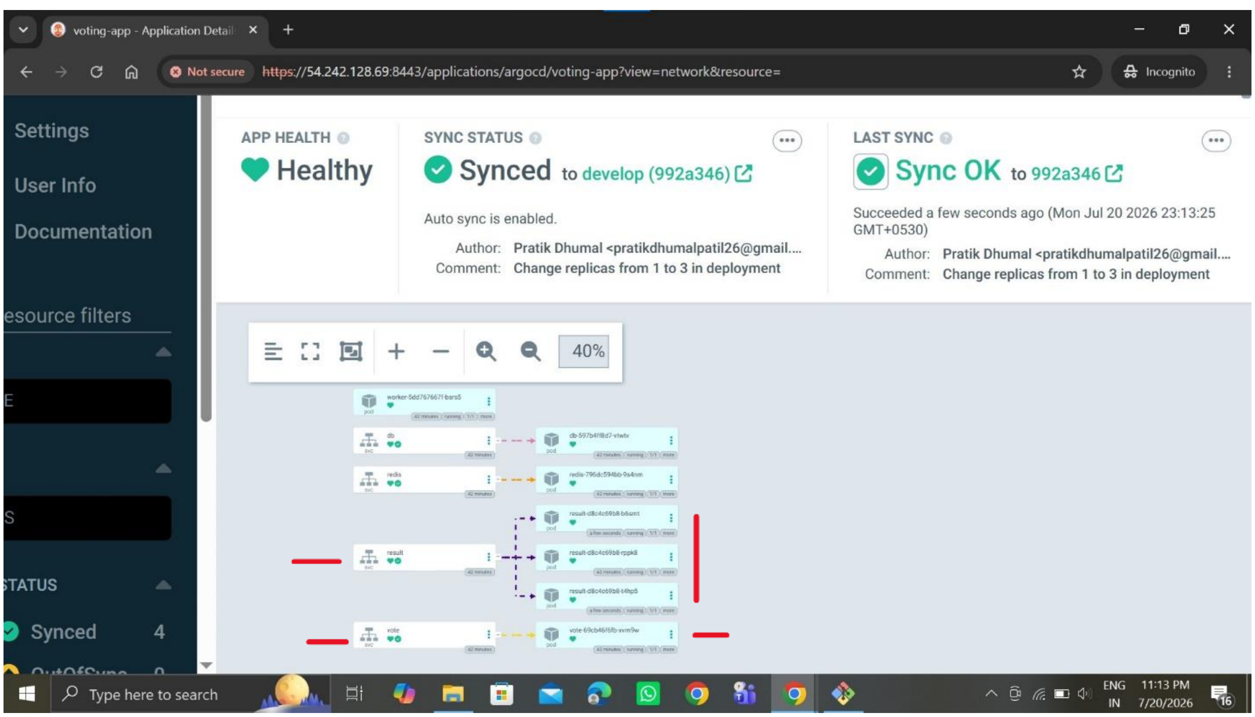
Then set vote replicas 2 to 1



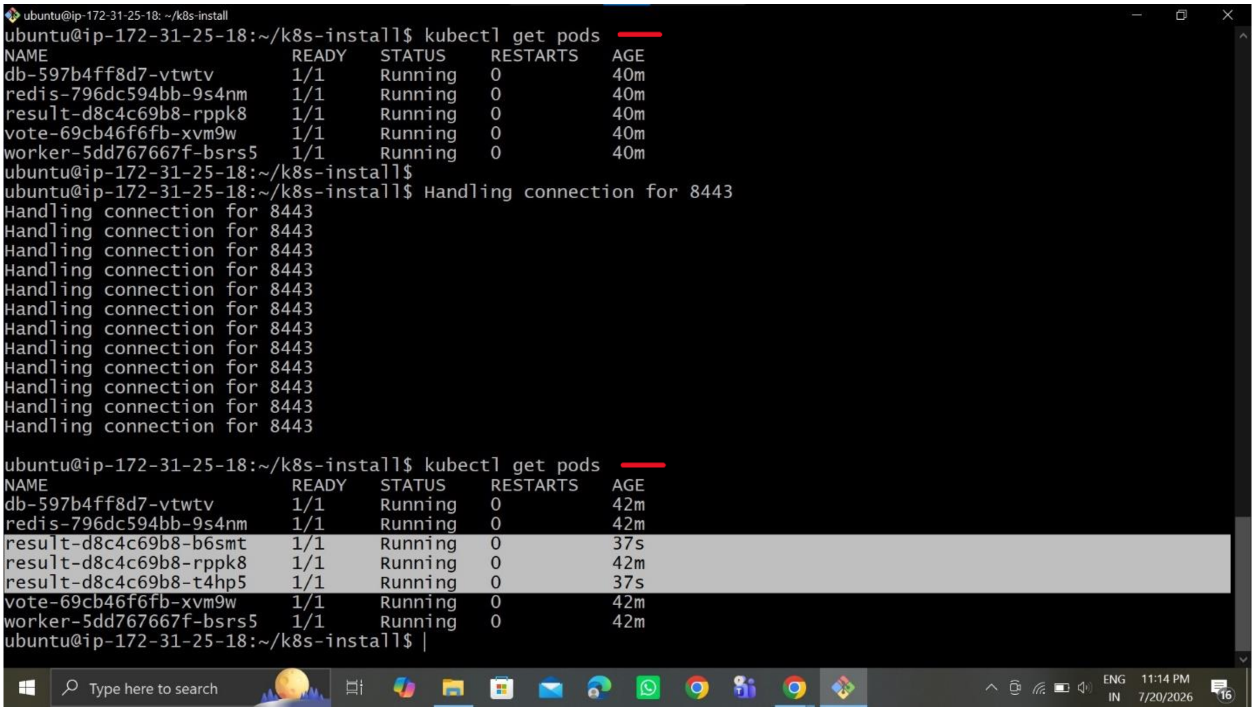
Set result replicas 1 to 3



Successfully show this pods vote 1 pods running & result 3 pods running

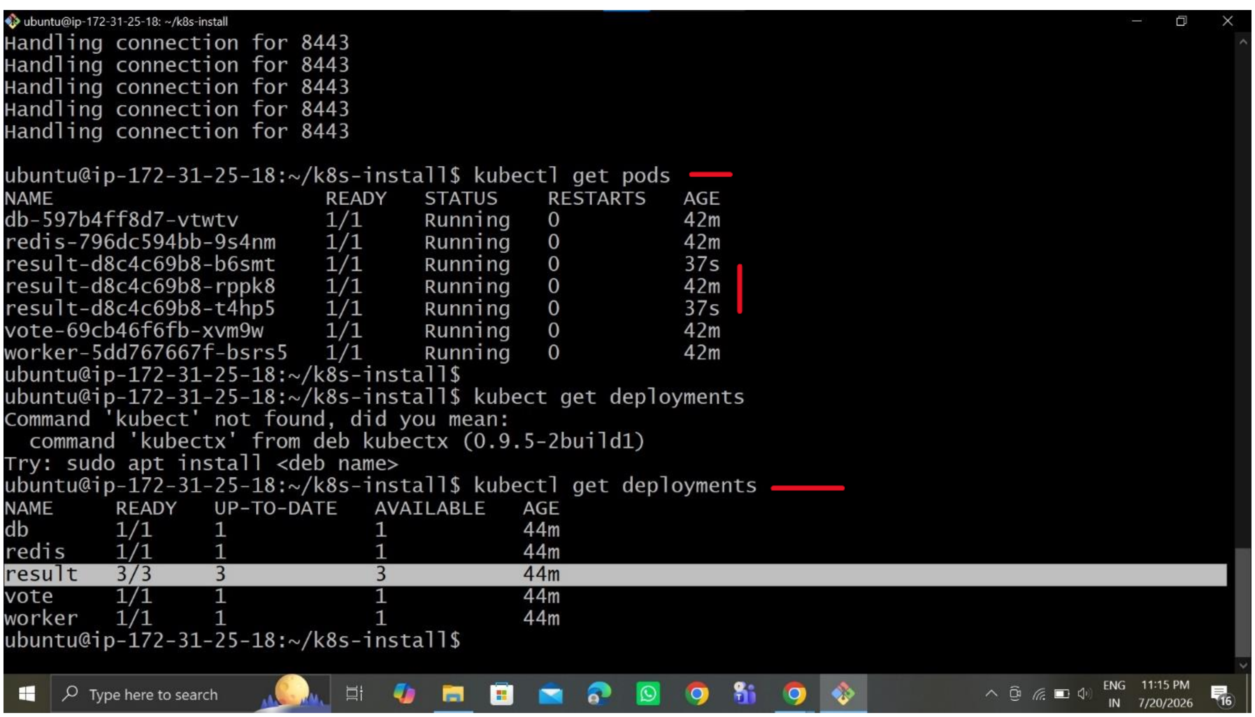


Successfully show this pods in terminal




```
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
NAME                       READY   STATUS    RESTARTS   AGE
db-597b4ff8d7-vtwtv        1/1     Running   0          40m
redis-796dc594bb-9s4nm     1/1     Running   0          40m
result-d8c4c69b8-rppk8     1/1     Running   0          40m
vote-69cb46f6fb-xvm9w      1/1     Running   0          40m
worker-5dd767667f-bsrs5    1/1     Running   0          40m
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443
Handling connection for 8443

ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
NAME                       READY   STATUS    RESTARTS   AGE
db-597b4ff8d7-vtwtv        1/1     Running   0          42m
redis-796dc594bb-9s4nm     1/1     Running   0          42m
result-d8c4c69b8-b6smt     1/1     Running   0          37s
result-d8c4c69b8-rppk8     1/1     Running   0          42m
result-d8c4c69b8-t4hp5     1/1     Running   0          37s
vote-69cb46f6fb-xvm9w      1/1     Running   0          42m
worker-5dd767667f-bsrs5    1/1     Running   0          42m
ubuntu@ip-172-31-25-18:~/k8s-install$ |
```

result running 3 pods




```
Handling connection for 8443
Handling connection for 8443
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ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get pods
NAME                       READY   STATUS    RESTARTS   AGE
db-597b4ff8d7-vtwtv        1/1     Running   0          42m
redis-796dc594bb-9s4nm     1/1     Running   0          42m
result-d8c4c69b8-b6smt     1/1     Running   0          37s
result-d8c4c69b8-rppk8     1/1     Running   0          42m
result-d8c4c69b8-t4hp5     1/1     Running   0          37s
vote-69cb46f6fb-xvm9w      1/1     Running   0          42m
worker-5dd767667f-bsrs5    1/1     Running   0          42m
ubuntu@ip-172-31-25-18:~/k8s-install$
ubuntu@ip-172-31-25-18:~/k8s-install$ kubect get deployments
Command 'kubect' not found, did you mean:
  command 'kubectx' from deb kubectx (0.9.5-2build1)
Try: sudo apt install <deb name>
ubuntu@ip-172-31-25-18:~/k8s-install$ kubectl get deployments
NAME     READY   UP-TO-DATE   AVAILABLE   AGE
db       1/1     1            1           44m
redis    1/1     1            1           44m
result   3/3     3            3           44m
vote     1/1     1            1           44m
worker   1/1     1            1           44m
ubuntu@ip-172-31-25-18:~/k8s-install$
```
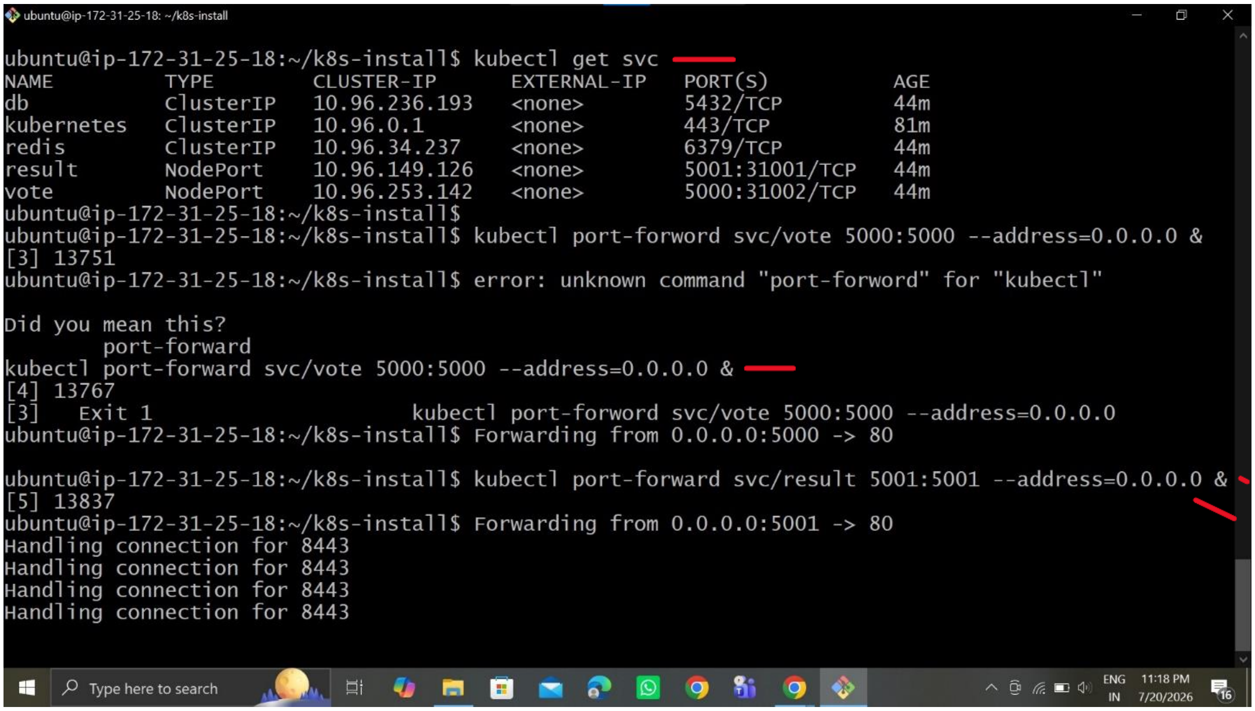
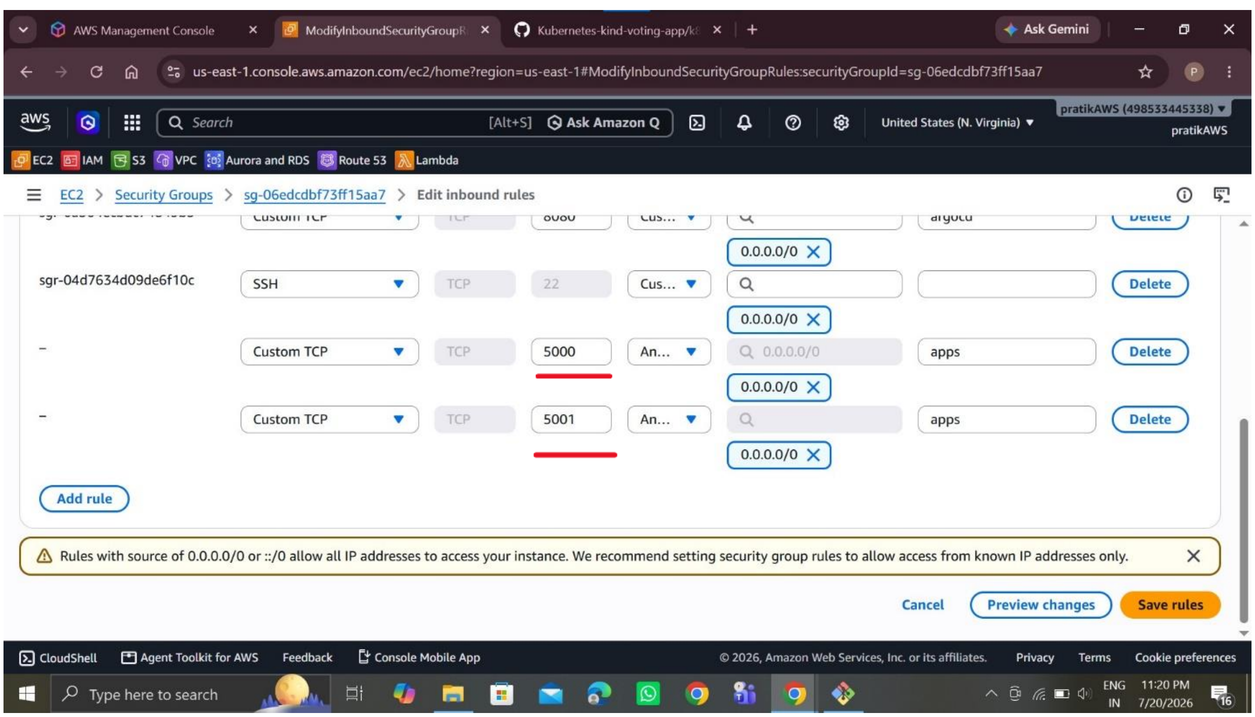
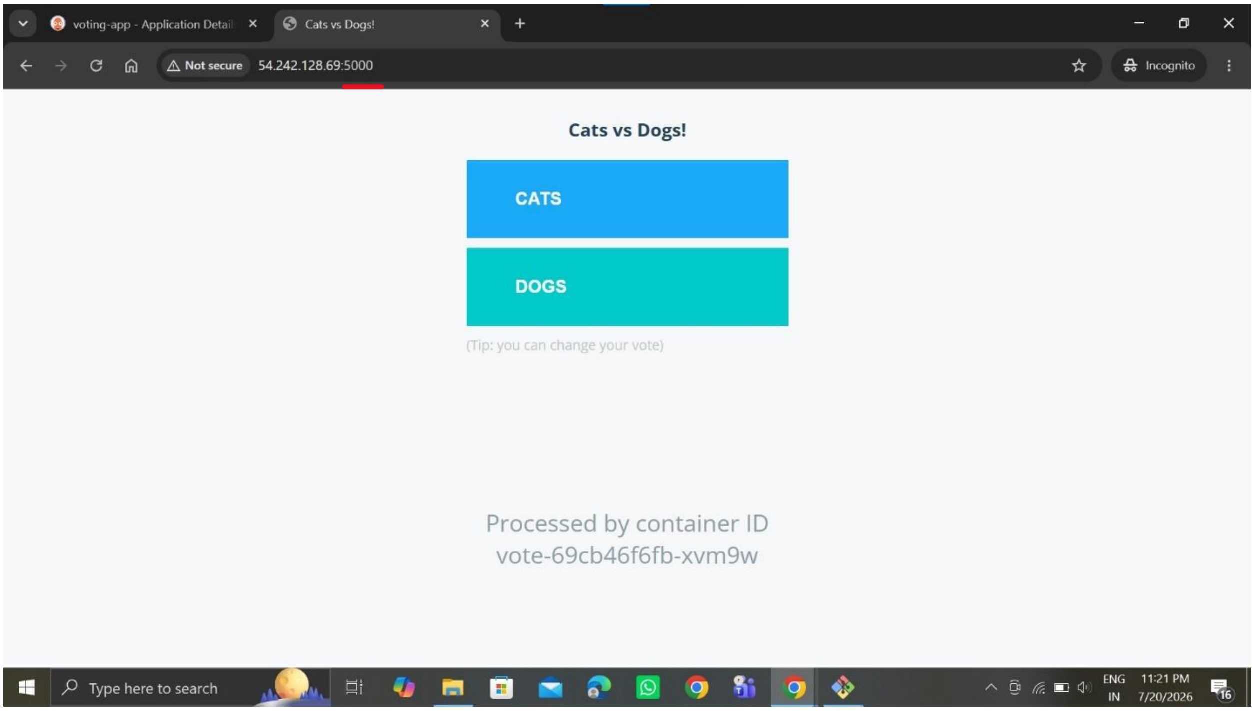

Port forward in vote & result



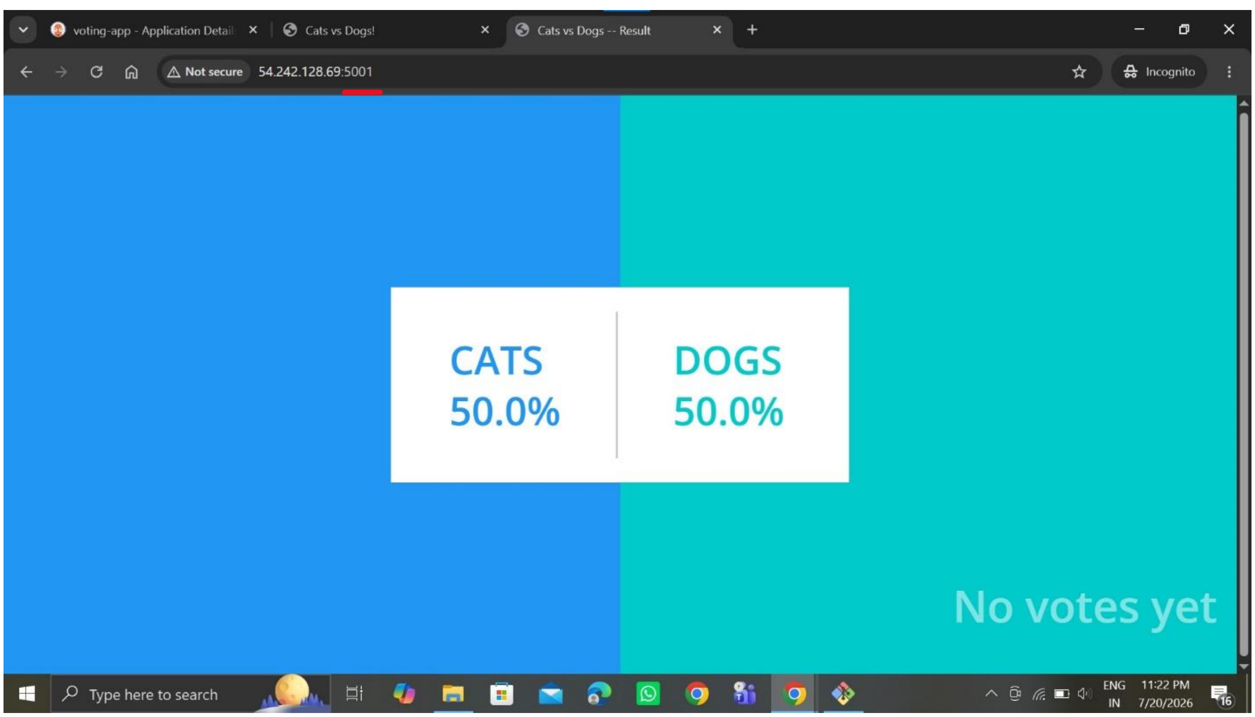
Allow this port in Security group



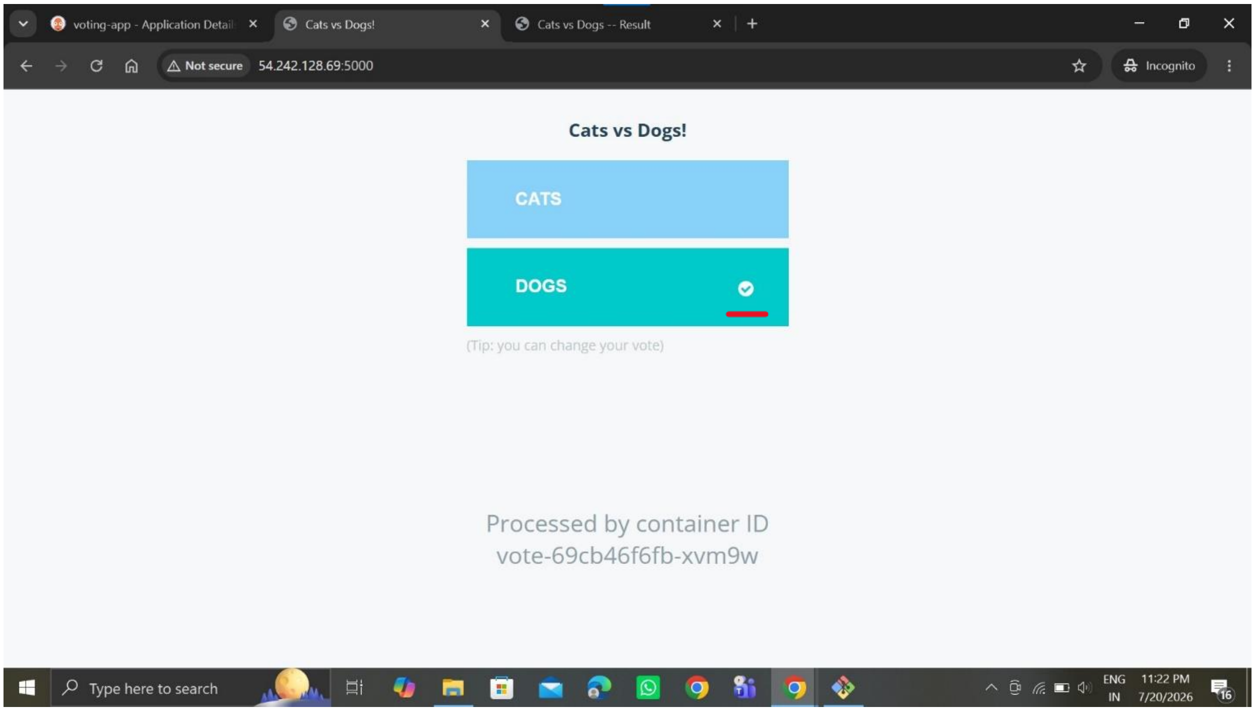
This is Vote page



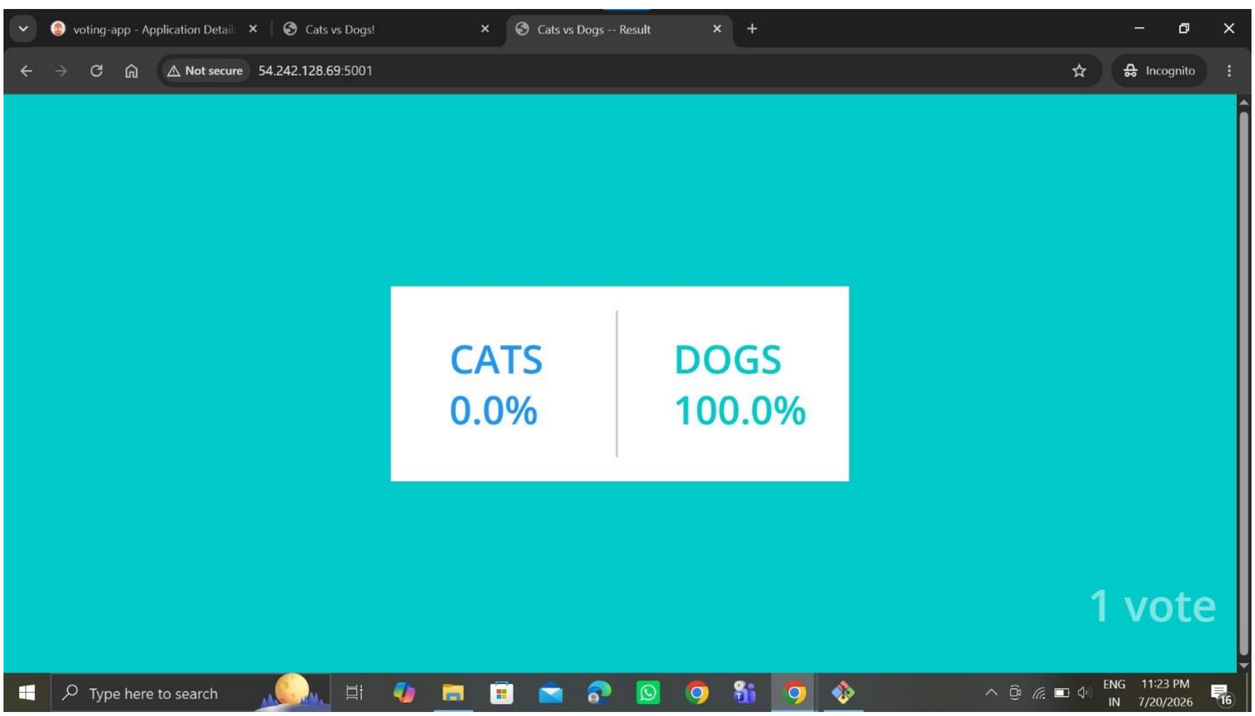
This is Result page



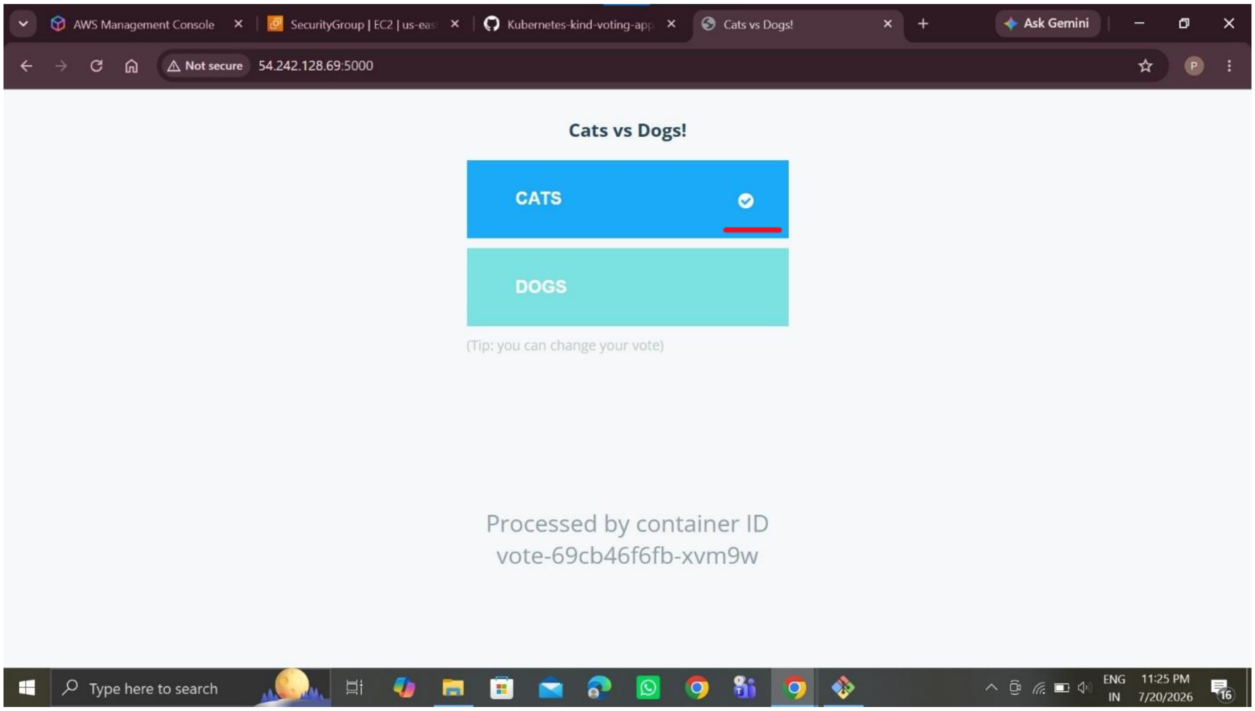
Then vote Dogs



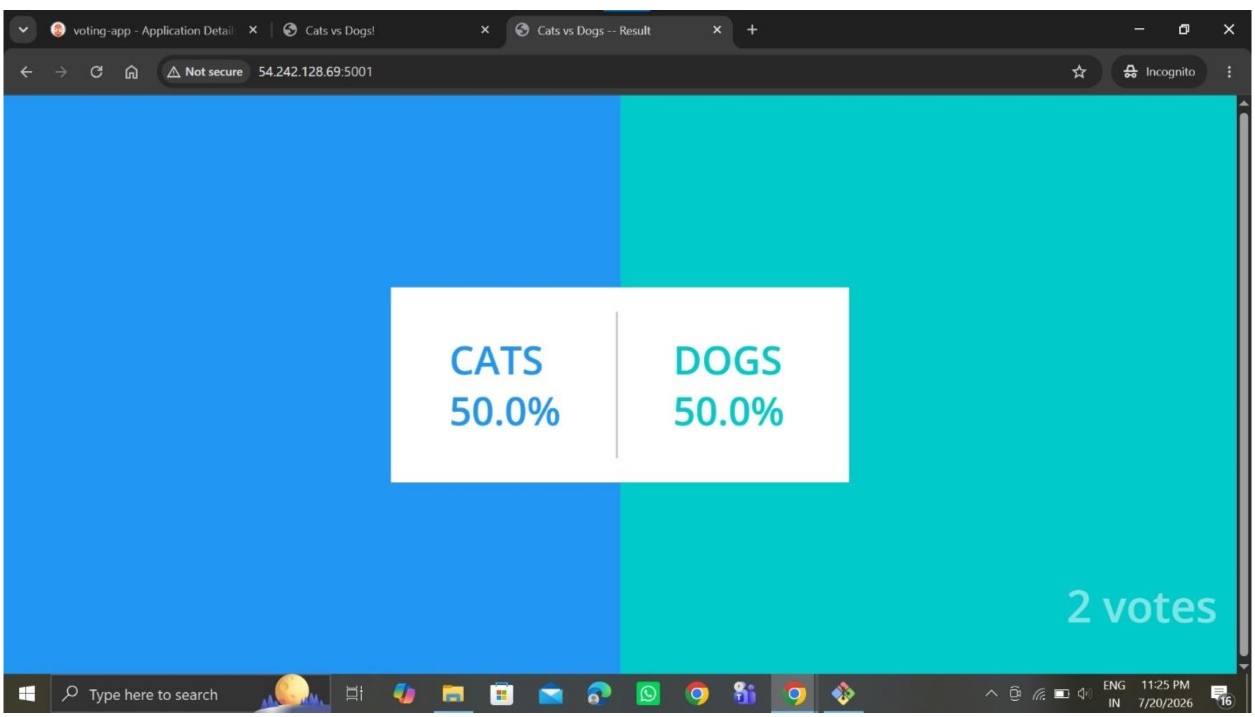
Result is 100% in Dogs



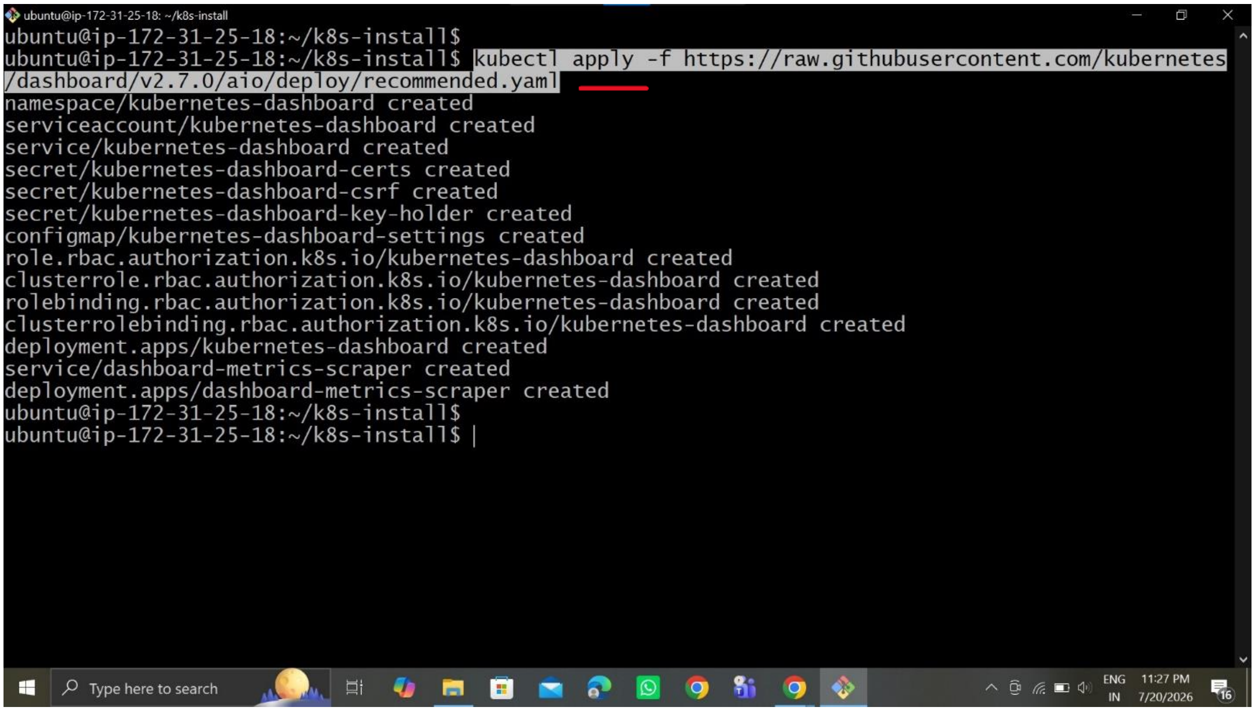
Then go to another browser tab and open vote page & vote to Cats



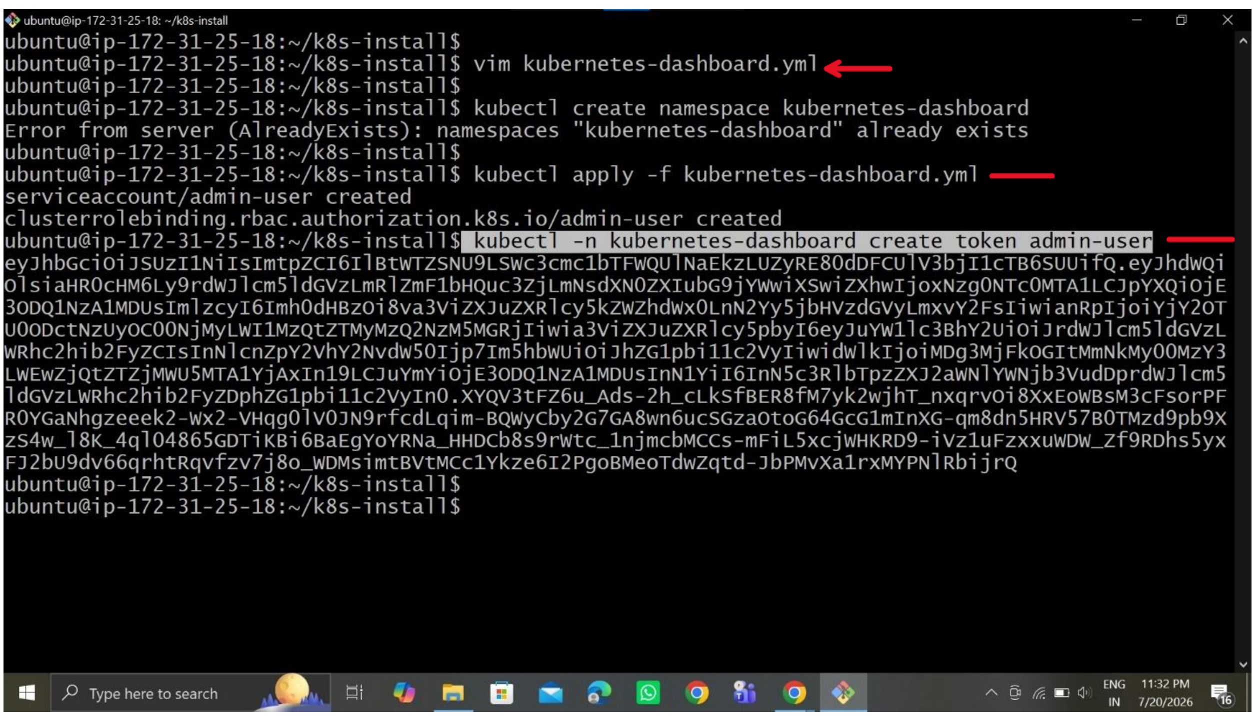
Then go to result page the result is equal



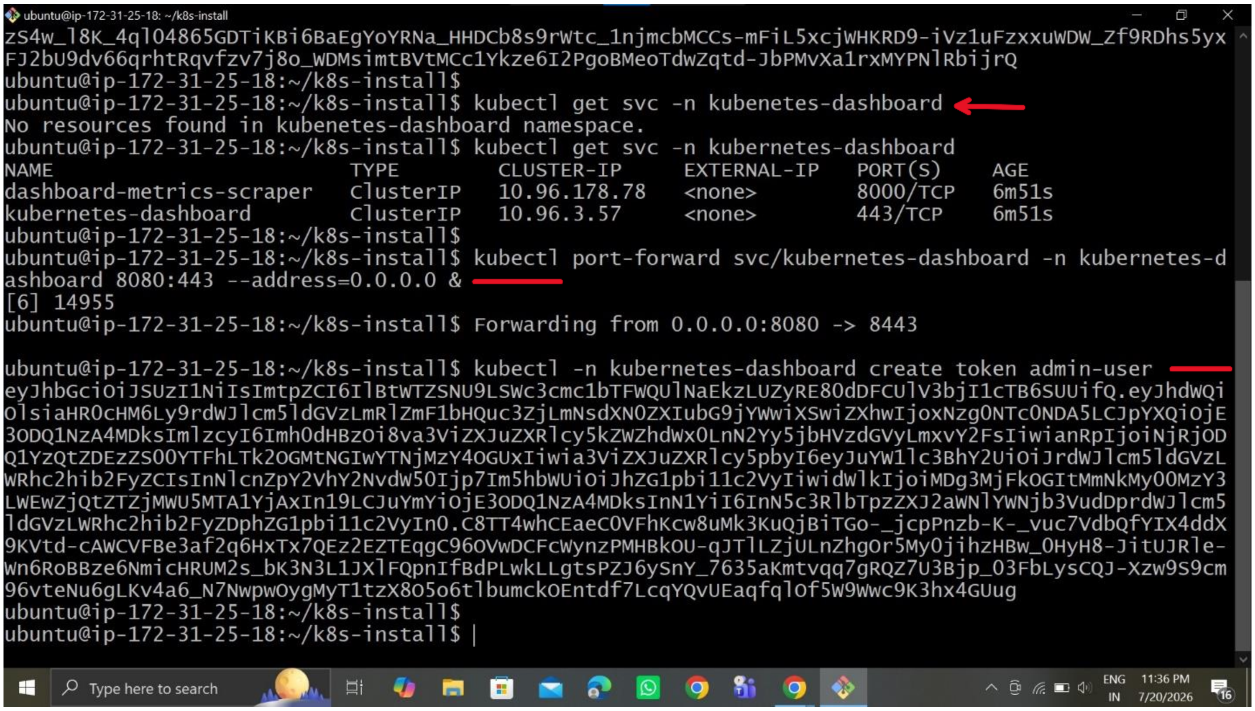
Deploy Kubernetes dashboard



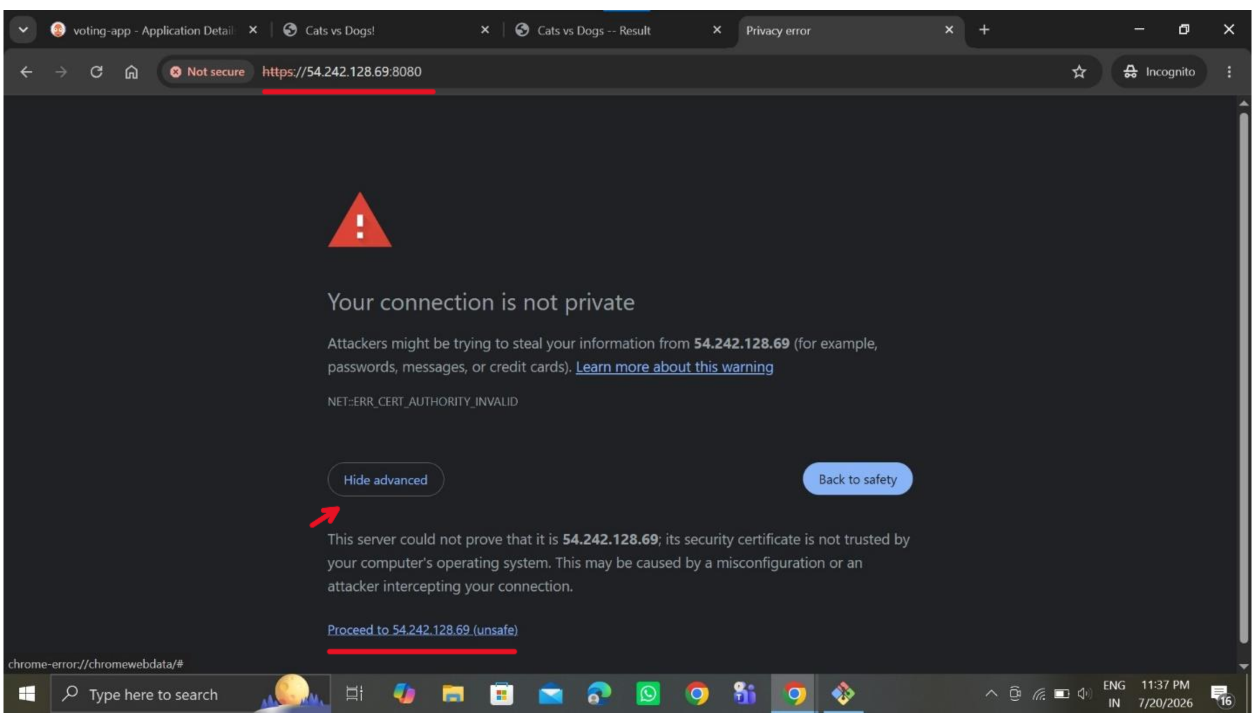
Then write dashboard .yml file & create admin-user , then apply & then generate Token



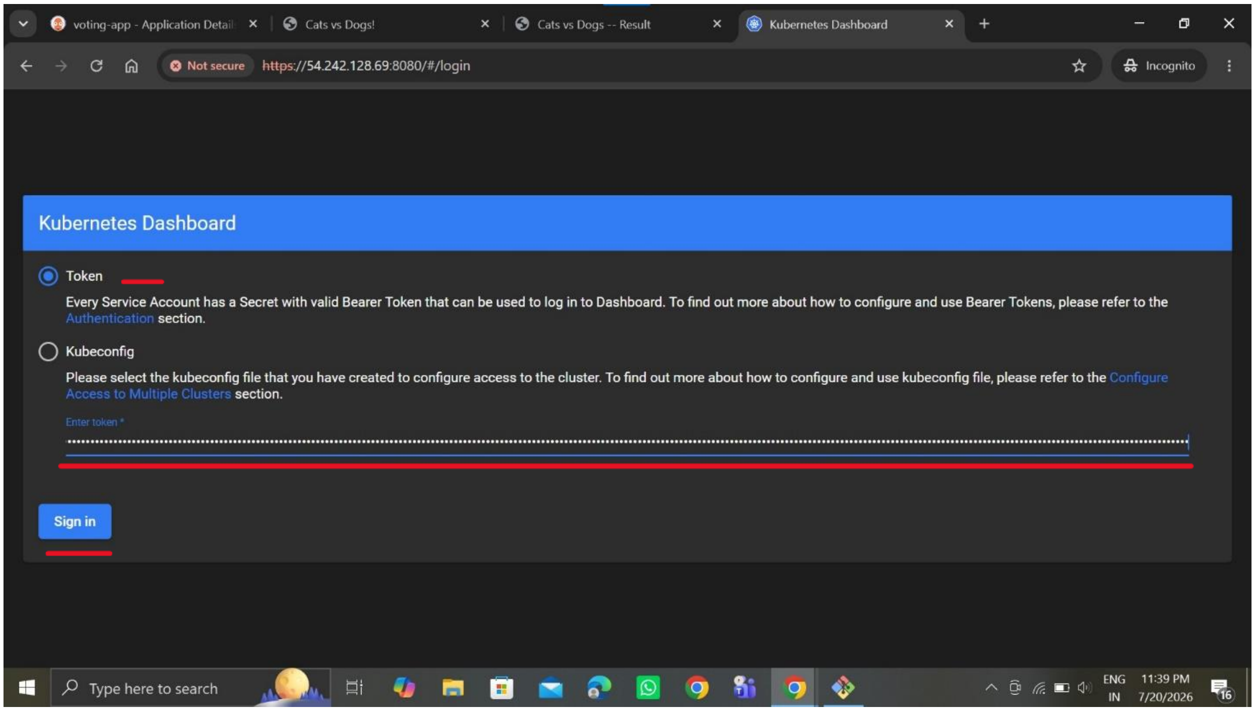
Then show Kubernetes dashboard service & then port forward in Kubernetes dashboard



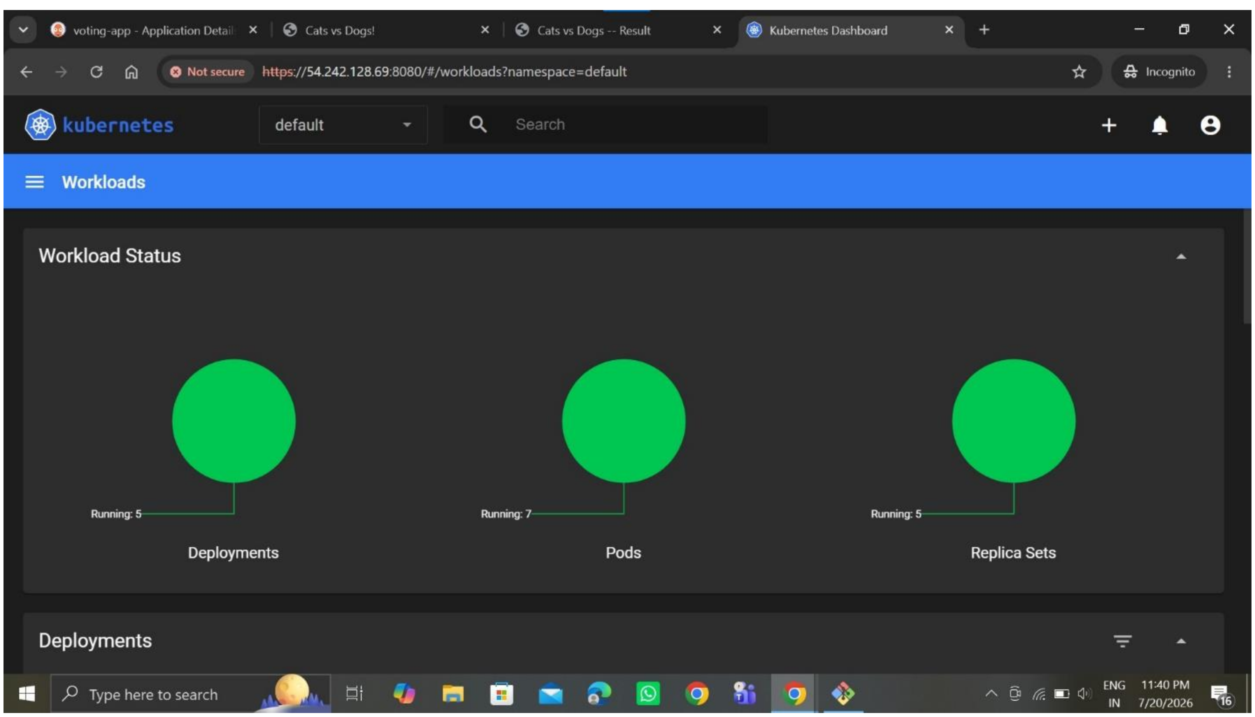
Then copy Public IP of Kubernetes cluster & go to browser & paste it



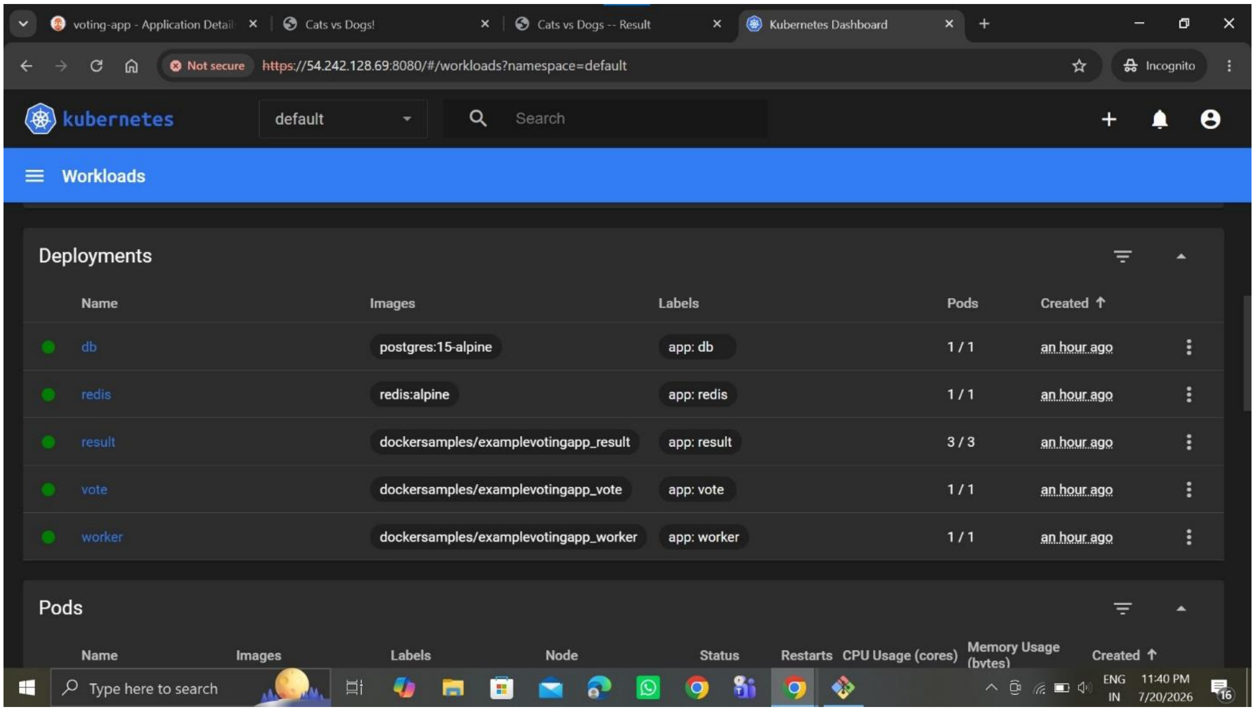
Then paste this generated Token & sign in



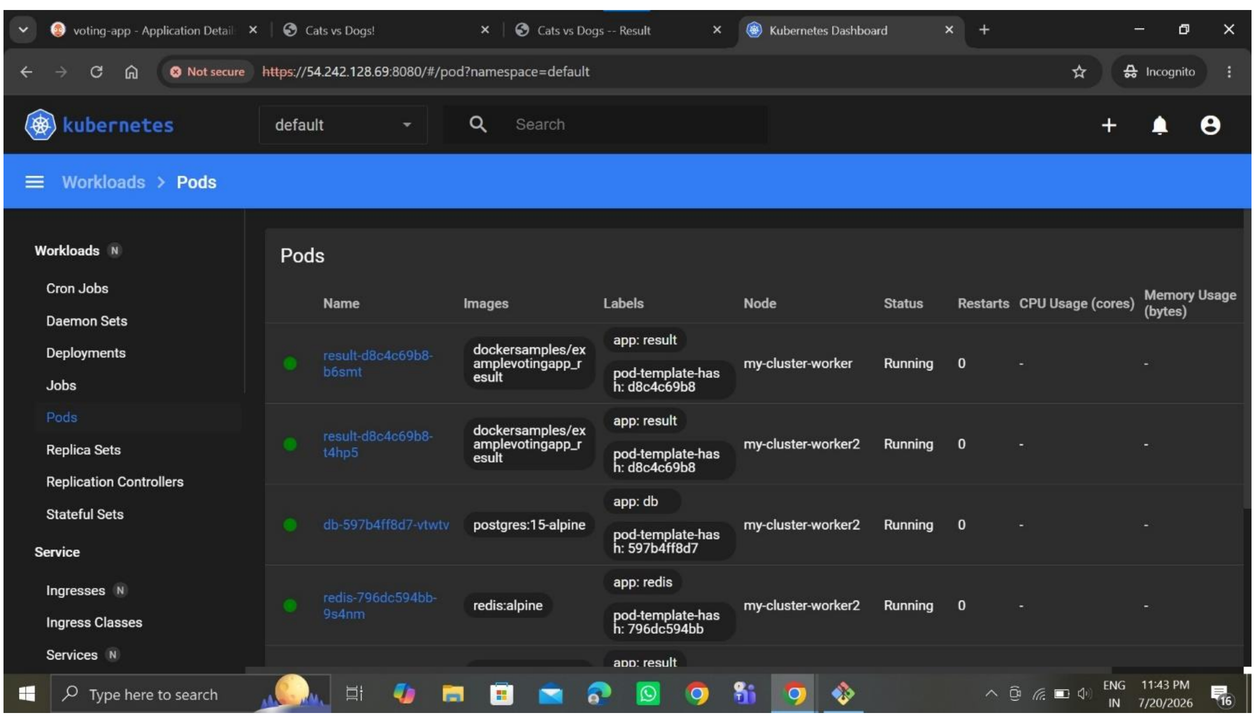
Successfully show the Kubernetes Dashboard

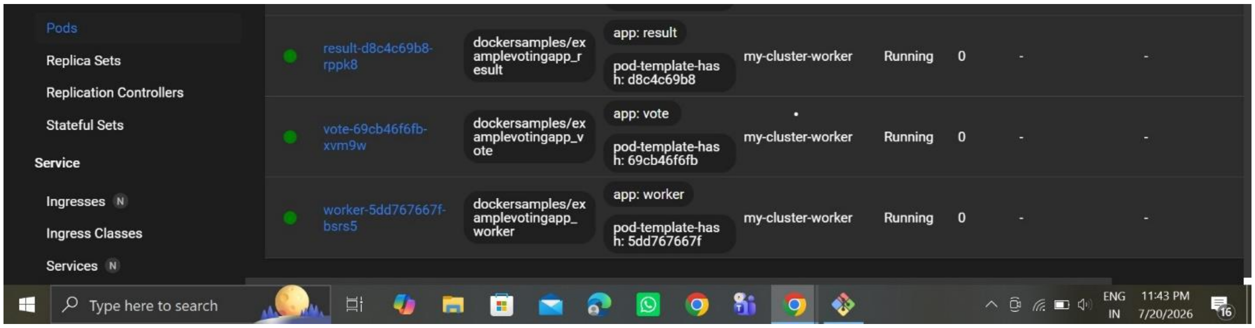


Deployments pods

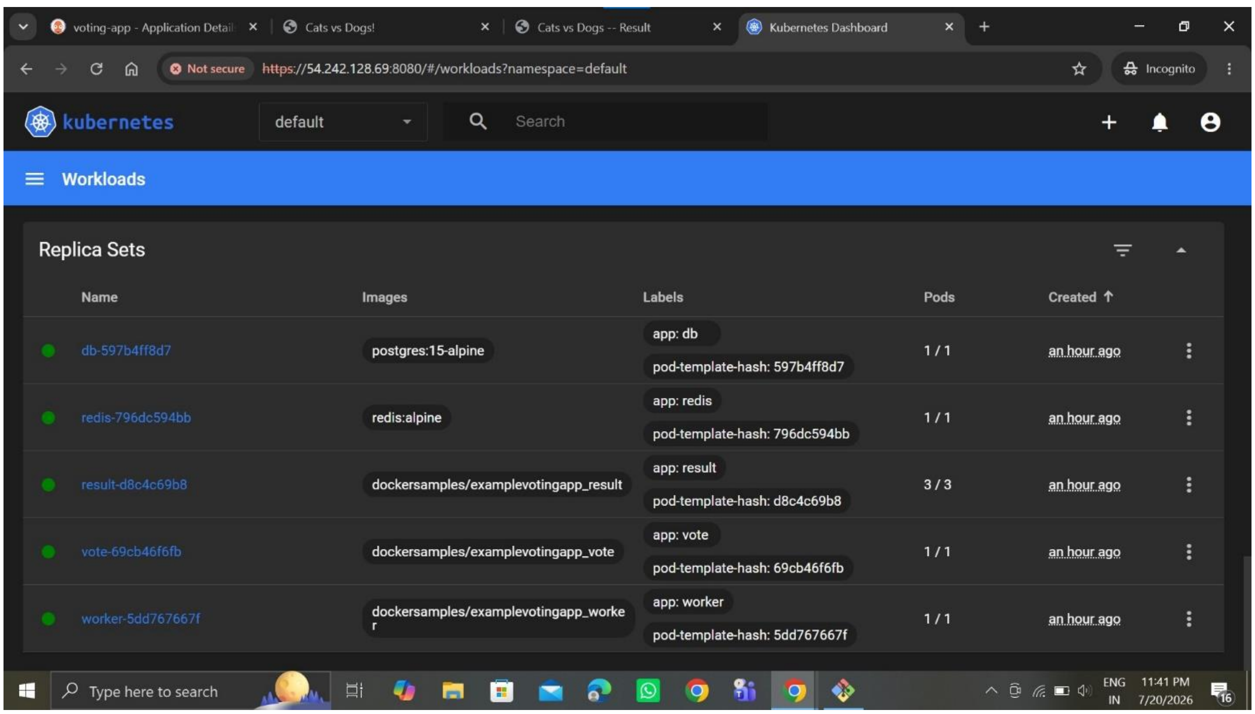


Pods





Replica sets



Automated Deployment of Scalable Applications on AWS EC2 with Kubernetes and Argo CD Successfully .